Database Search

Contents

Searching in Databases

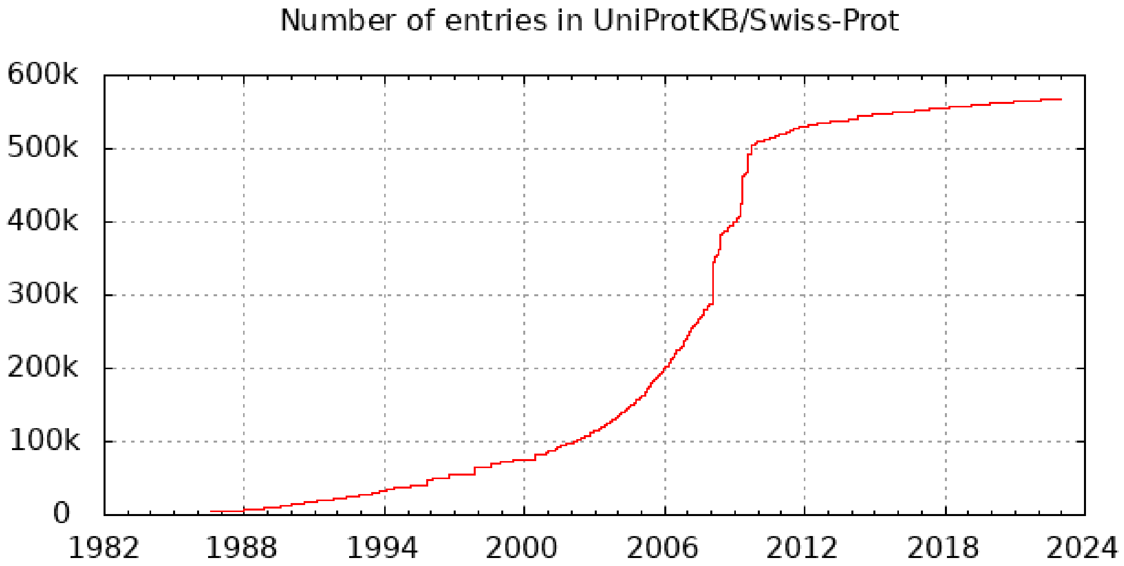
Searching in databases I – Reason

- When working with new sequences we need to gather as much information about our sequence as possible
- To do so we need to compare it to already known and analyzed sequences from databases like the UniProtKB/Swiss-Prot
- Therefore we need to look for similar / homologous sequences in these databases with the help of specialized algorithms

Searching in databases II – Principle

- Differs fundamentally from alignments: you search for possible homolog sequences
- Not only best matches but also similar matches need to be found
- Algorithms like Smith-Waterman or ClustalW are too slow
- Idea: many sequences of a database don’t match the query
- Filter all sequences that aren’t similar enough (reduction of the data)
- Most known:
  - BLAST
  - FastA

Searching in databases III – Entries in Uniprot/Swiss-Prot [1]



- If we look at the entries in the UniProtKB/Swiss-Prot we realize the dimensions of this small database
- Other databases like the non-redundant protein database of the NCBI or the non-redundant UniProtKB contain even millions of sequences

Large-Scale Lemma

Large-Scale Lemma I

- When working with these database search methods we need to figure out whether or not a sequence we found is a relevant hit
- The problem is that in a never-ending, randomly generated text, each arbitrary substring can be guaranteed to be found

→ That means if our sequence database is big enough we can find each arbitrary sequence with a great probability

→ This is called the Large-Scale Lemma

Large-Scale Lemma II – E-Value [8]

- In general the E-Value represents the anticipated number of alignments with the same score as your result in a random database
- The E-value is dependent on the length of the query sequence and the size of the database
- For example, an alignment obtaining an E-value of 0.05 means that there is a 5 in 100 chance of it occurring by chance alone
- You can find a mathematical explanation under [8]

Large-Scale Lemma III – Meaning of E-value [9]

- E-value < 10e-100 ➔ Identical sequences
  - You will get long alignments across the entire query and hit sequence
- 10e-100 < E-value < 10e-50 ➔ Almost identical sequences
  - A long stretch of the query protein is matched to the database
- 10e-50 < E-value < 10e-10 ➔ Closely related sequences
  - could be a domain match or similar
- 10e-10 < E-value < 1 ➔ potential homologue sequences but it is a gray area
- E-value > 1 ➔ Proteins are most likely not related
- E-value > 10 ➔ Hits are most likely junk unless the query sequence is very short

FastA

FastA I [2]

- FastA is an algorithm that was developed by David J. Lipman and William R. Pearson in 1985 [3]
- FastA takes a given nucleotide or amino acid sequence and searches a corresponding sequence database by using local sequence alignments to find matches of similar database sequences
- The FastA program follows a largely heuristic method which contributes to the high speed of its execution
- It’s the namesake of the fasta format which is omnipresent in bioinformatics, especially when working with sequence data

FastA II [4]

- The algorithm contains the following steps:
  - Split up the query sequence in k-mers (k=2 for proteins and k=4-6 for nucleotides), small sequence fragments
  - Search for regions in the database sequences that are identical with the k-mers of the query, to find identical fragments
  - Find the best regions containing such fragments and score them according to a substitution matrix
  - Try to combine these identical fragments by adding gaps in between them to form an approximate Alignment
  - If these approximate alignments score better than a given threshold, an optimal pair-wise Alignment is calculated and handed over as result

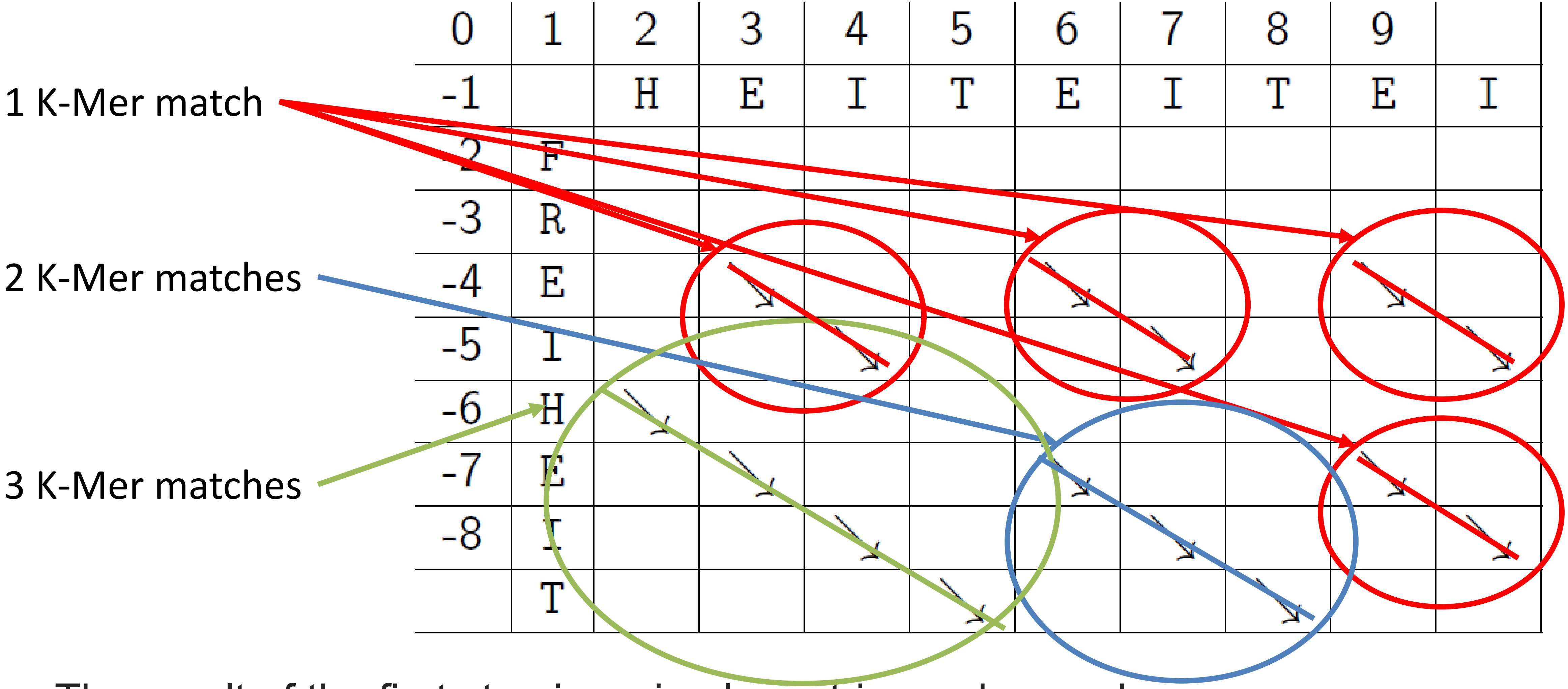
FastA III – Step 1

- In this first step our search sequence, also called query sequence, is split up into so-called k-mers
  - These are small fragments of our sequence consisting of:
    - 1 to 2 amino acids
    - 4 to 6 nucleotides
- These k-mers are overlapping with each other:
  - For example the sequence ACTG produces the three 2-mers AC, CT and TG
- Afterwards the database is searched for appearances of these k-mers in other sequences

FastA IV – Step 1

- For each database sequence all Hits of k-mers are marked in am alignment matrix as small diagonals
- Subsequent Hits lead to longer diagonals, depending on the number of hits that can be found in direct neighborhood to each other
- Only sequences that have enough Hits and thus diagonals that are long enough can be considered going into the next step
- For each of these sequences the 10 best diagonals, the longest ones, are saved and taken over to the next step
- In this step FastA only looks for matches and thus does NOT use any form of subsitution matrix

FastA V – Step 1



- The result of the first step is a simple matrix as shown above

FastA VI – Step 2

- In the second step, the ten regions that were saved in step 1 are analyzed again
- They are rescored using BLOSUM 50 or another matrix depending on the parameters you define
    - That means the different match scores of these diagonals are added together
- The diagonal that produces the best score is called Init 1 and the values of these Init 1 regions are used to rank all found sequences up to this point

FastA VII – Step 2

Scores: 11 (6 + 5)
E/E + I/I

Score: 16 (6 + 5 + 5)
E/E + I/I + T/T

Score: 26 (10 + 6 + 5 +5)
H/H + E/E + I/I + T/T

0	1	2	3	4	5	6	7	8	9	
-1		H	E	I	T	E	I	T	E	I
-2	F									
-3	R									
-4	E									
-5	I									
-6	H									
-7	E									
-8	I									
	T									

- Looking back at our Example, the different diagonals produce scores of 11, 16 and 26 respectively)

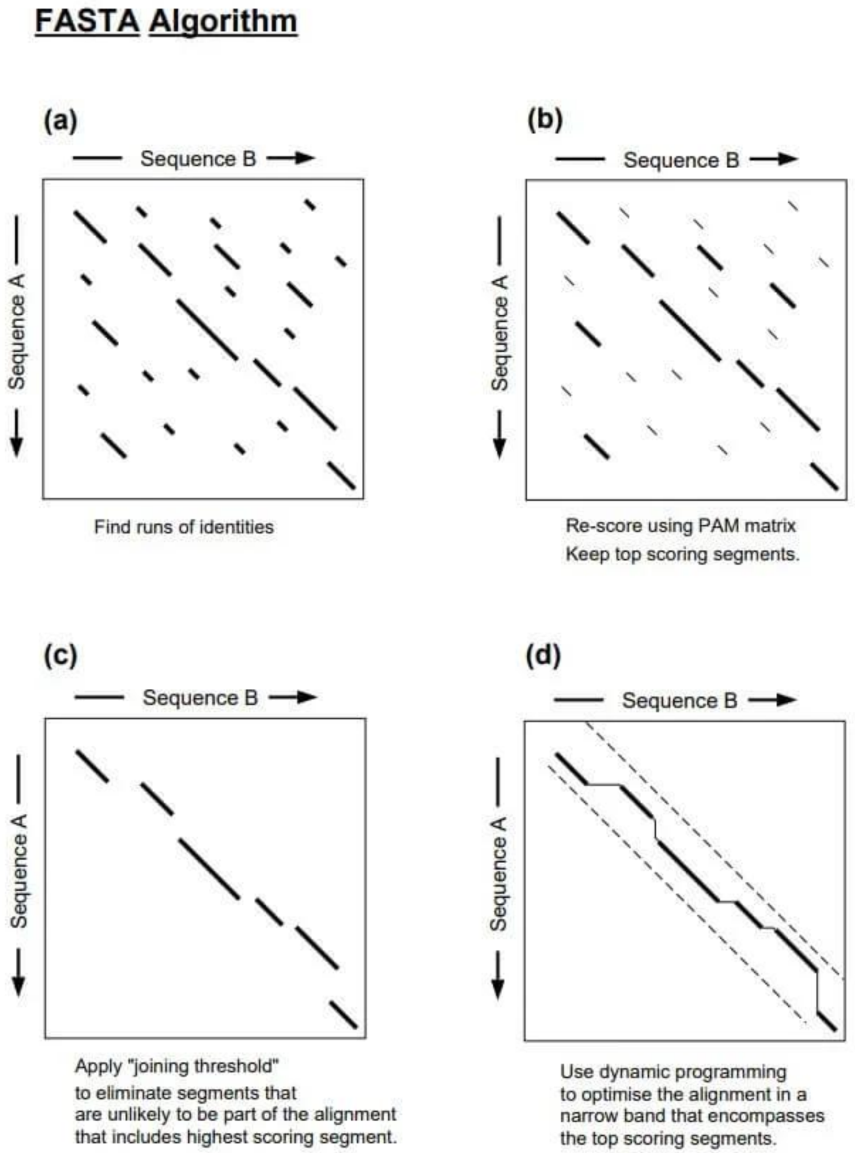
FastA VIII – Step 3

- In the third step the high scoring regions found in the first two steps are combined to form an approximate alignment alongside the main diagonal of the matrix and including the Init 1 region of step 2
- This is done by simply introducing gaps in between the regions and penalizing them relatively harsh
  - Per default a gap-open penalty of -10 and a gap-extension penalty of -2 is used
- Only approximate Alignments that score better than a Threshold are taken into account as Hits and are handed over to the last step

FastA IX – Step 4

- In the last step optimal Alignments between the query sequence and the found sequences from the database are calculated
- An adjusted version of the Smith-Waterman algorithm is used to optimize the time needed
- Simply said the algorithm takes the already identified diagonals and creates the alignment around them
- The number of results depends on the settings you choose, per default we get back the top 50 sequences

FastA X – Summary [5]



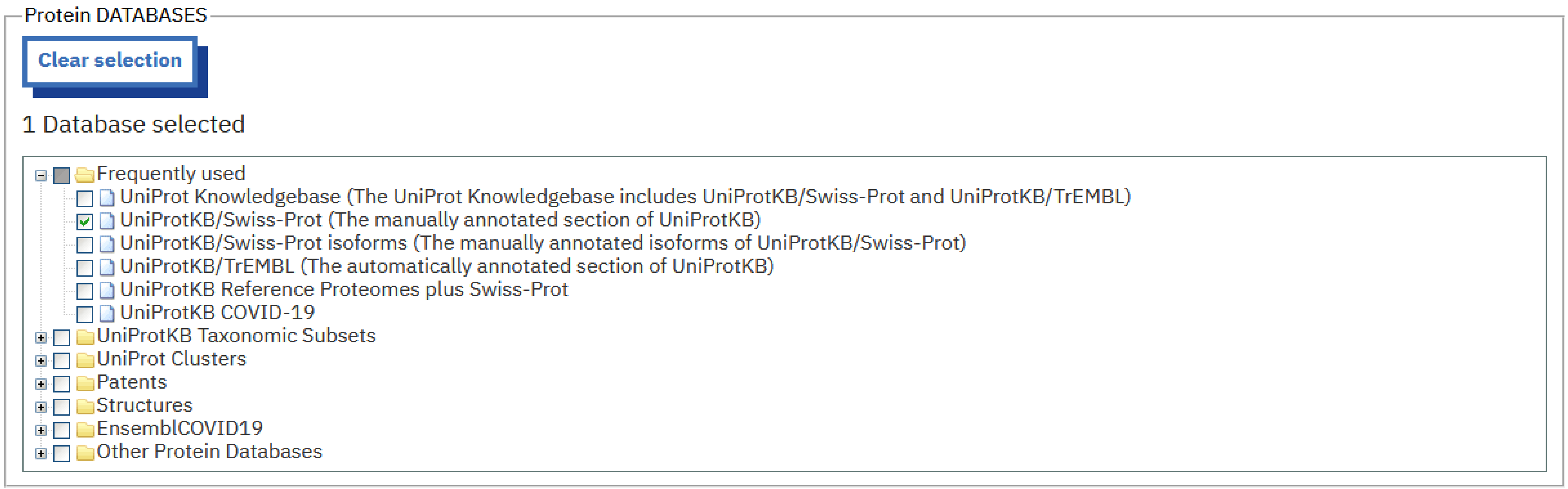
- On the left side you can see a graphical summary of the algorithm
  a. Searching for k-mer hits
  b. Scoring diagonals using BLOSUM50 for example
  c. Creating an approximate alignment of the found diagonals
  d. Calculating the optimal Alignment using the diagonals as basis

FastA XI – Web service

- We can use the FastA algorithm using a web service provided by the EMBL:
  - https://www.ebi.ac.uk/jdispatcher/sss/fasta
- The web page allows us to adjust the search by:
  - Using different databases
  - Adjusting the search parameters
  - Adjusting the number of results
  - Using different Algorithms

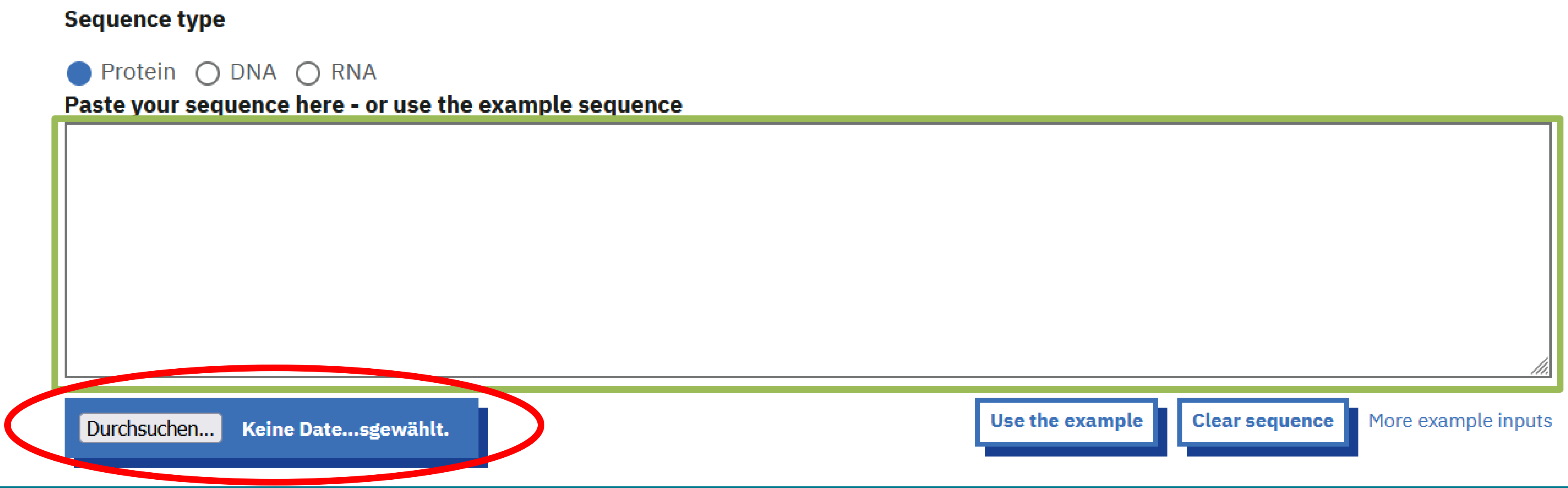
FastA XII – Web service – Databases

- At the top we are able to choose in which database we want to search
- Per default the UniProtKB/SwissProt is selected and you always should start your search in there
- If the results are not good enough you can repeat the search using the UniProt Knowledgebase



FastA XIII – Web service – Input

- In the next step you can either paste your sequence into the editor (green rectangle)
- You can also upload a fasta file (red circle)
- However in both cases we can use ONLY ONE sequence at a time



FastA XIV – Web service – Program choice

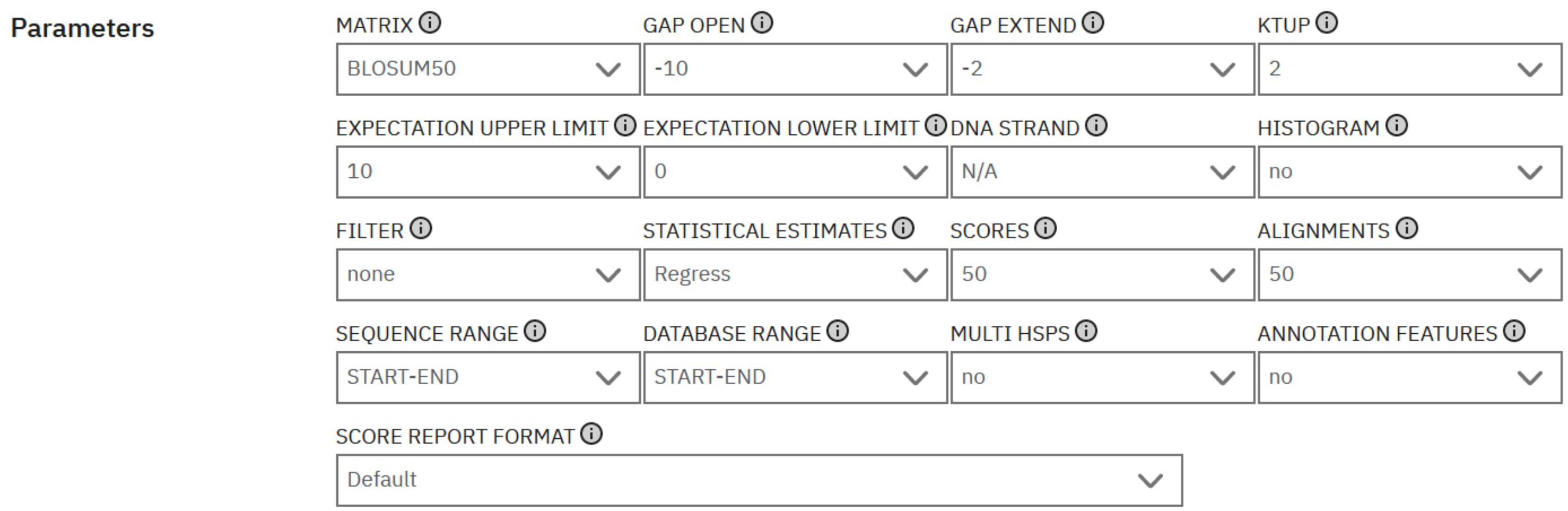
- After you entered your sequence you can specify which program you want to use
- Per default the FastA algorithm is selected
- The other options are FastX, FastY, Ssearch, GGSearch and GLSearch

FastA XV – Web service – Default Parameters

- You can adjust the parameters by clicking on More options (red circle)

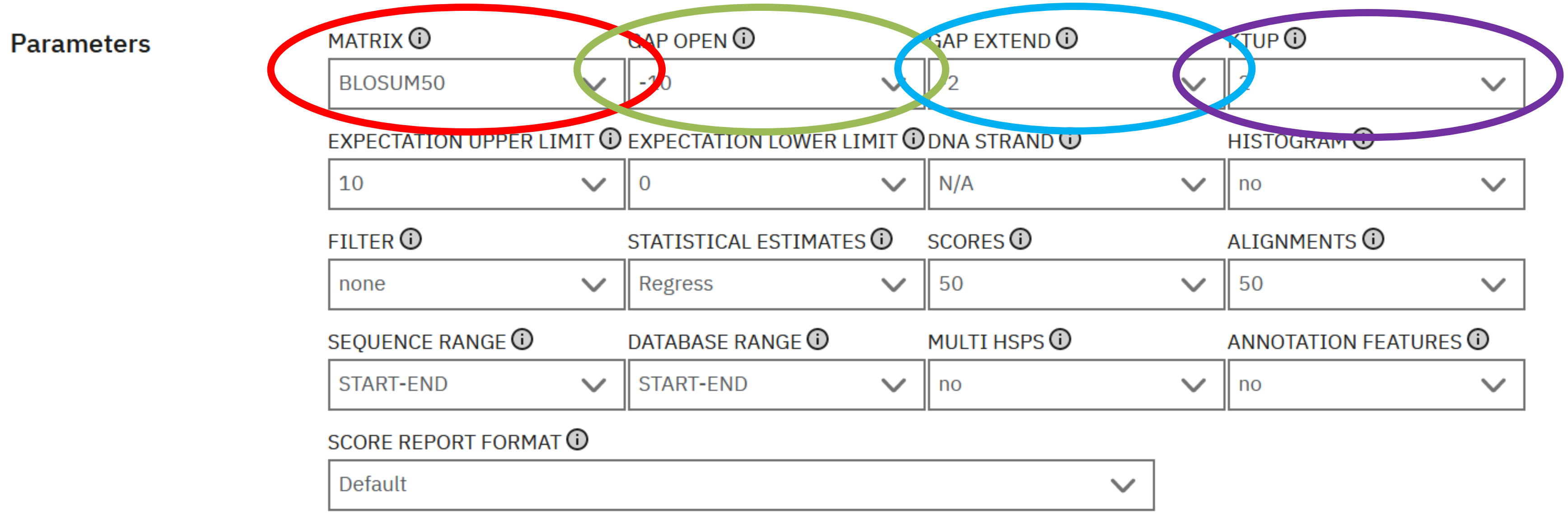
FastA XVII – Web service – Adjusting parameters

- There are 17 different parameters we can adjust when using the web service
- If you‘re interested in a bit more in depth explanation about them you can use the help page of the EMBL:
  - https://www.ebi.ac.uk/seqdb/confluence/display/JDSAT/FASTA+Help+and+Documentation



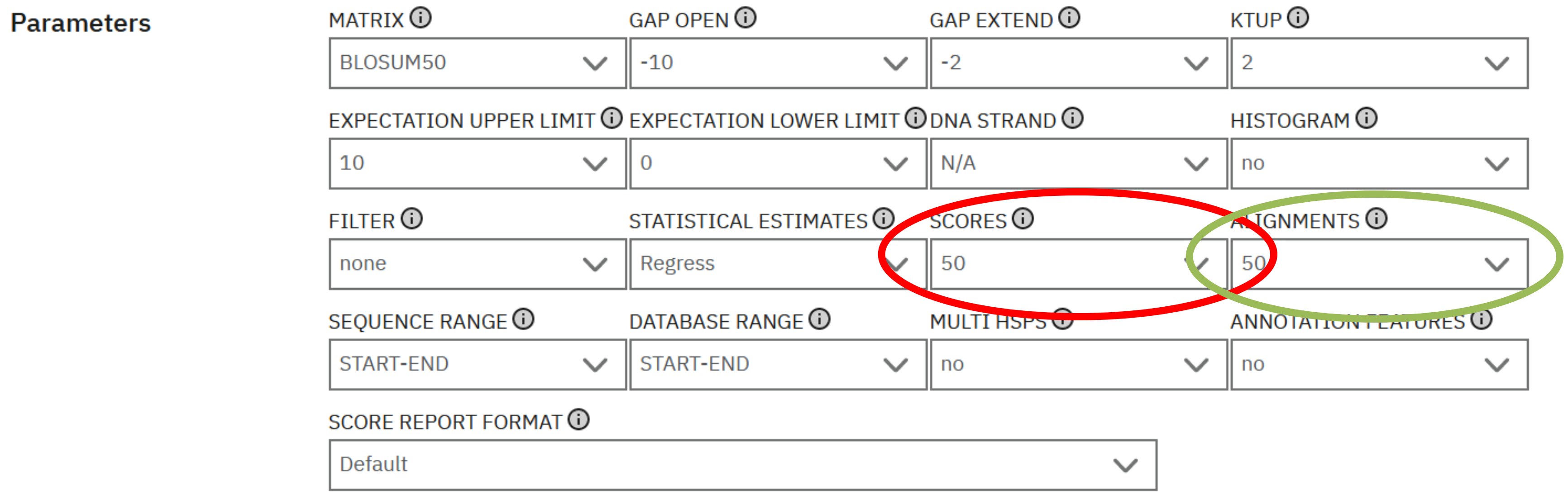
FastA XVIII – Web service – Adjusting parameters

- For us the most interesting parameters are:
  - MATRIX ➔ defines which substitution matrix you use (red)
  - GAP OPEN ➔ Refers to the gap-open penalty (green)
  - GAP EXTEND ➔ Refers to the gap-extend penalty (blue)
  - KTUP ➔ Refers to the size of the k-mer (purple)



FastA XIX – Web service – Adjusting parameters

- For us the most interesting parameters are:
  - SCORES ➔ Refers to the maximal amount of results presented (red)
  - ALIGNMENTS ➔ Refers to the number of optimal alignments calculated for the results, need to be equal or smaller than the number of scores (green)



FastA XX – Web service – Start search

- Once you‘re finished with setting up your search you can start the program by clicking on submit (red circle)
- If you want you can specify a Title for your Job

FastA XXI – Web service – Result page I

- Once your search is finished you reach the results page
- On it you have a tabular overview of all found sequences (right)
- Furthermore you can look at additional results using the tabs at the top (below)

14 results found

Items per page: 50 1 – 14 of 14

☑	Align	DB:ID	Source	Length	Score(Bits)	Identities(%)	Positives(%)	E()
☑	1	SP:O48788	Probable inactive receptor kinase At2g26730 OS=Arabidopsis thaliana OX=3702 GN=At2g26730 PE=1 SV=1 **View Cross-references** [10] ▶ Gene expression ▶ Bioactive molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Literature ▶ Samples & ontologies ▶ Molecular interactions ▶ Protein families ▶ Protein expression data ▶ Protein sequences	658	40.0	29.1	59.6	0.47
☑	2	SP:O48788	Probable inactive receptor kinase At2g26730 OS=Arabidopsis thaliana OX=3702 GN=At2g26730 PE=1 SV=1 **View Cross-references** [10] ▶ Gene expression ▶ Bioactive molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Literature ▶ Samples & ontologies ▶ Molecular interactions ▶ Protein families ▶ Protein expression data ▶ Protein sequences	658	40.0	29.1	59.6	0.47
☑	3	SP:Q5AAU3	Protein transport protein SEC31 OS=Candida albicans (strain SC5314 / ATCC MYA-2876) OX=237561 GN=PGA63 PE=3 SV=2 **View Cross-references** [7] ▶ Bioactive molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Literature ▶ Samples & ontologies ▶ Protein families ▶ Protein sequences	1265	39.0	22.8	53.0	1.8
☑	4	SP:Q5AAU3	Protein transport protein SEC31 OS=Candida albicans (strain SC5314 / ATCC MYA-2876) OX=237561 GN=PGA63 PE=3 SV=2 **View Cross-references** [7] ▶ Bioactive molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Literature ▶ Samples & ontologies ▶ Protein families ▶ Protein sequences	1265	39.0	22.8	53.0	1.8

Summary Table   Tool Output   Visual Output   Functional Predictions   Result Files   Submission Details

FastA XXII – Web service – Result page II

- Under “Tool Output” you can see the detailed outputs the program produces for your search
- This includes a list of the best scores found in the database
- The number of scores displayed here depends on the number you chose earlier
- You can download the results as an xml-file by clicking on download

```
# /fasta/bin/fasta36 -l /prod/fastacfg/fasta3db -L -T 32 -p -s BL50 -r +5/-4 -f -10 -g -2 -E "10.0 -1.0" -F 0.0 -b 50 -d 50 -m "F9B fasta-
I20240125-102401-0099-11685745-p1m.m9" -m "F10 fasta-I20240125-102401-0099-11685745-p1m.m10" -z 1 @:1- +uniprotkb_swissprot+ 2
FASTA searches a protein or DNA sequence data bank
 version 36.3.8h May, 2020
Please cite:
 W.R. Pearson & D.J. Lipman PNAS (1988) 85:2444-2448

Query: @
  1>>>tr|B3JI28|B3JI28_9BACT Uncharacterized protein OS=Phocaeicola coprocola DSM 17136 OX=470145 GN=BACCOP_01540 PE=1 SV=1 - 608 aa
Library: UniProtKB
  511313063 residues in 1304960 sequences

Statistics:  Expectation_n fit: rho(ln(x))= 8.3211+/-0.000141; mu= 5.8931+/- 0.008
 mean_var=63.7020+/-12.450, 0's: 527 Z-trim(113.0): 648  B-trim: 1059 in 1/63
 Lambda= 0.160693
 statistics sampled from 60000 (64300) to 125339 sequences
Algorithm: FASTA (3.8 Nov 2011) [optimized]
Parameters: DNA matrix (15:-5), open/ext: -10/-2
 ktup: 2, E-join: 1 (0.363), E-opt: 0.2 (0.0962), width:  16
 Scan time: 51.600

The best scores are:                                      opt bits E(1304960)
SP:O48788 Y2267_ARATH Probable inactive receptor k ( 658)  134 40.0    0.47
SP:O48788 Y2267_ARATH Probable inactive receptor k ( 658)  134 40.0    0.47
SP:Q5AAU3 SEC31_CANAL Protein transport protein SE (1265)  131 39.0     1.8
SP:Q5AAU3 SEC31_CANAL Protein transport protein SE (1265)  131 39.0     1.8
SP:P81575 CUPA1_CANPG Cuticle protein AM/CP1114 OS ( 102)  106 34.4     3.5
SP:P81575 CUPA1_CANPG Cuticle protein AM/CP1114 OS ( 102)  106 34.4     3.5
SP:P42255 ABFB_ASPNG Alpha-L-arabinofuranosidase B ( 499)  118 36.4     4.3
SP:A2R511 ABFB_ASPNC Probable alpha-L-arabinofuran ( 499)  118 36.4     4.3
SP:P42255 ABFB_ASPNG Alpha-L-arabinofuranosidase B ( 499)  118 36.4     4.3
SP:A2R511 ABFB_ASPNC Probable alpha-L-arabinofuran ( 499)  118 36.4     4.3
SP:Q8NK89 ABFB_ASPKW Alpha-L-arabinofuranosidase B ( 499)  113 35.3     9.6
SP:Q9C4B1 ABFB_ASPAW Probable alpha-L-arabinofuran ( 499)  113 35.3     9.6
SP:Q8NK89 ABFB_ASPKW Alpha-L-arabinofuranosidase B ( 499)  113 35.3     9.6
SP:Q9C4B1 ABFB_ASPAW Probable alpha-L-arabinofuran ( 499)  113 35.3     9.6
```

Tool output

Download (XML)

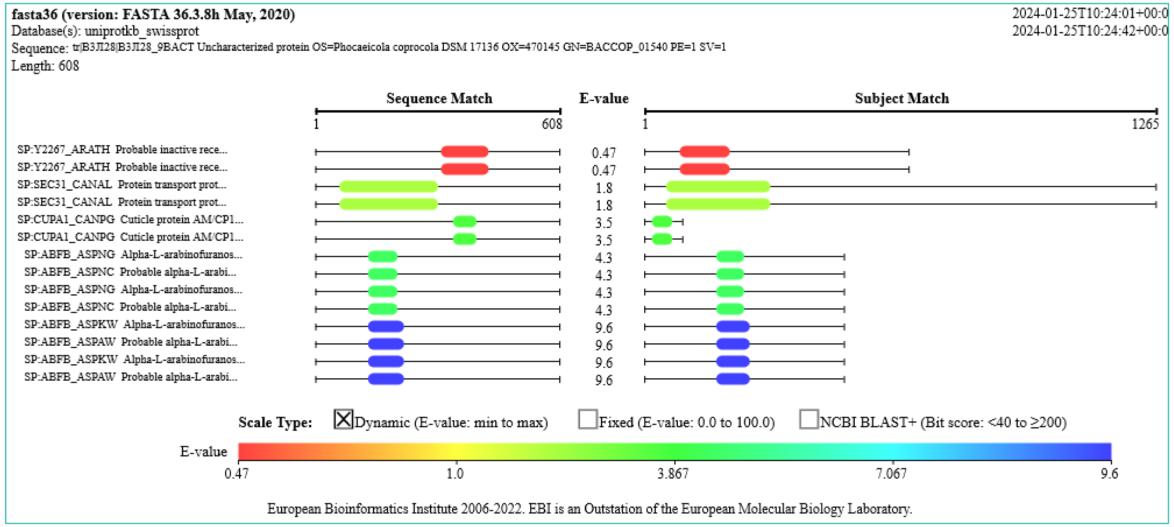
FastA XXIII – Web service – Result page III

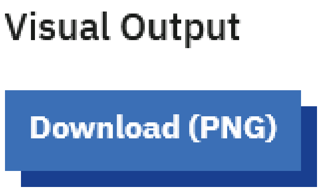
- If you chose to display a Histogram of your results you will get a histogram of Z-Scores for your results under “Tool Output”
- A Z-Score or standard score is generally calculated as follows [6]:
  - $Z = \dfrac{X - \mu}{\sigma}$
  - X is the calculated value, μ the mean value and σ the standard deviation
- For FastA the calculation is a bit different as described under [7]

```
         opt      E()
< 40   1070      0:======
  42      2      0:=          one = represents 210 library sequences
  44     13      2:*
  46     99     22:*
  48    438    141:*==
  50   1106    573:==*===
  52   2127   1620:=======*===
  54   3420   3435:================*
  56   5707   5821:===========================*
  58   6928   8280:=================================      *
  60   9922  10276:================================================*
  62  11507  11470:======================================================*
  64  12570  11791:========================================================*===
  66  12219  11369:======================================================*====
  68   9983  10431:================================================ *
  70   8739   9210:========================================== *
  72   7582   7893:=====================================*
  74   6260   6610:============================== *
  76   5036   5438:======================== *
  78   4070   4412:==================== *
  80   3201   3541:================*
  82   2630   2819:=============*
  84   2254   2229:==========*
  86   1733   1755:========*
  88   1298   1376:======*
  90   1081   1075:=====*
  92    864    839:===*=
  94    964    653:===*=
  96    657    507:==*=
  98    467    394:=*=
 100    337    301:=*
 102    261    239:=*
 104    231    185:*=
 106    162    143:*
```

FastA XXIV – Web service – Result page IV

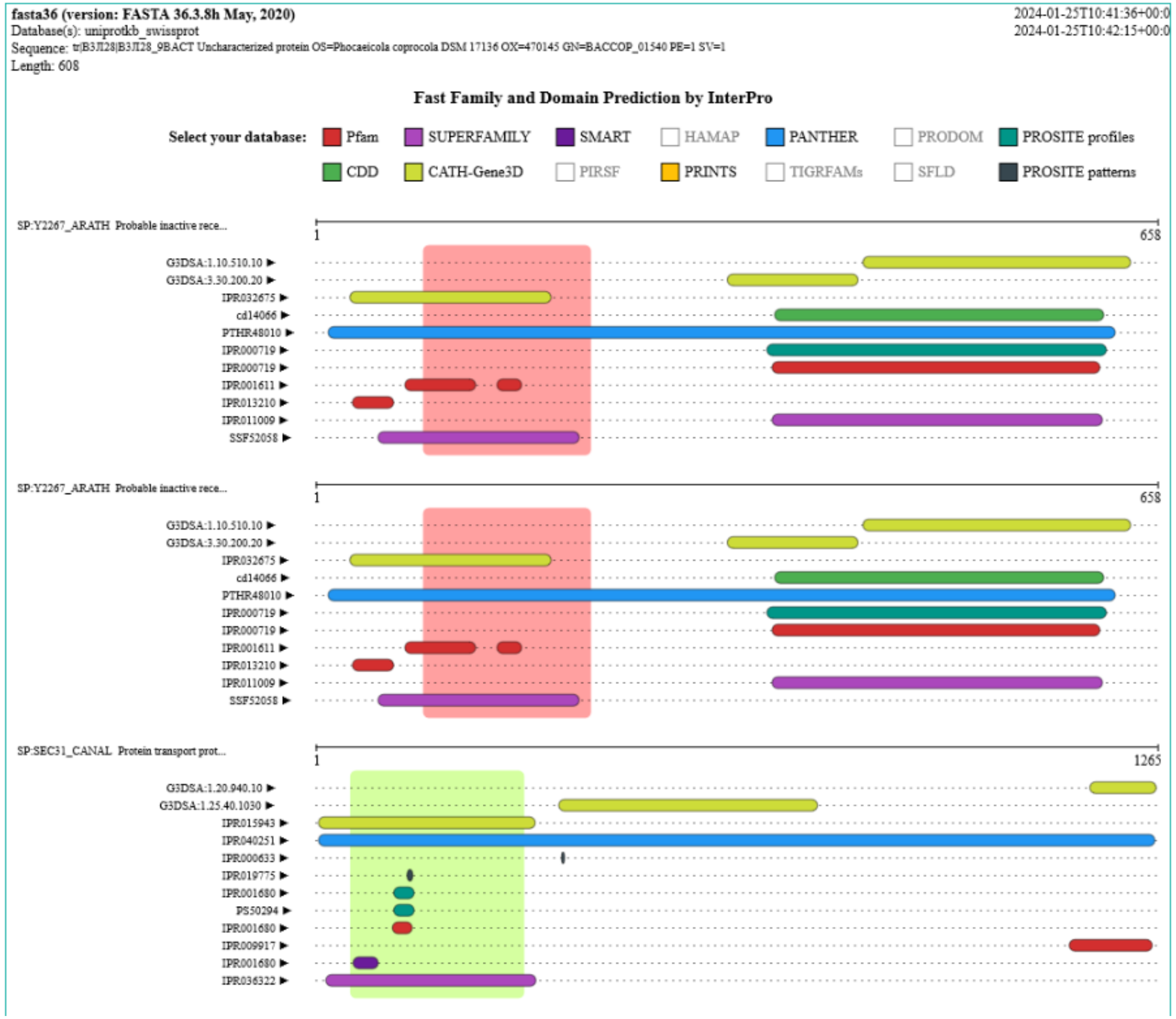
- Under “Visual Output” you can get a graphic representation of your alignments
- The left side shows the part of your query sequence that was aligned and the right side the part of the database sequence that is part of the resulting alignment
- You can download the whole image as .png file





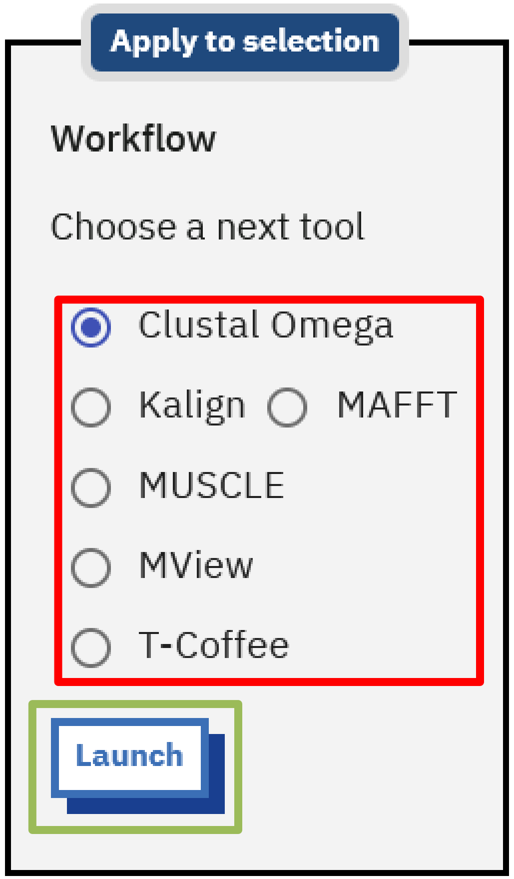
FastA XXV – Web service – Result page V

- Under “Functional Predictions” you can see a graphical overview of (potential) functions of the sequences you found in the database
- These stem from different secondary databases like GENE3D, PANTHER, PFAM, PIRSF, PRINTS, PRODOM, PROFILE, PROSITE, SMART, SSF and TIGERFAMs



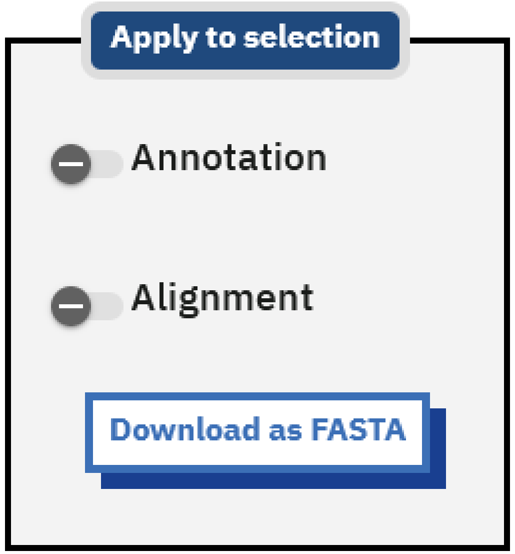
FastA XXVI – Web service – Result page VI

- On the left side of the "Summary Table" you have a Sidebar that gives you access to some more options
- At the bottom you are able to create a MSA out of all sequences you found using different Tools (image)
- You can choose between Clustal Omega, Kalign, MAFFT, MUSCLE, MVIEW and T-Coffee (red rectangle)
- Afterwards by clicking on Launch a new tab is opened with the corresponding tool and your sequences already in the entry field (green rectangle)



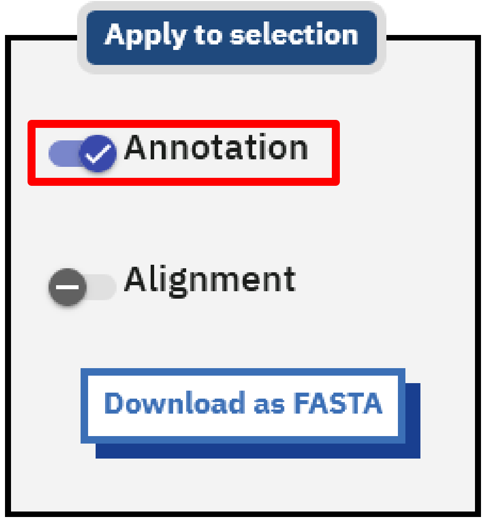
FastA XXVII – Web service – Result page VII

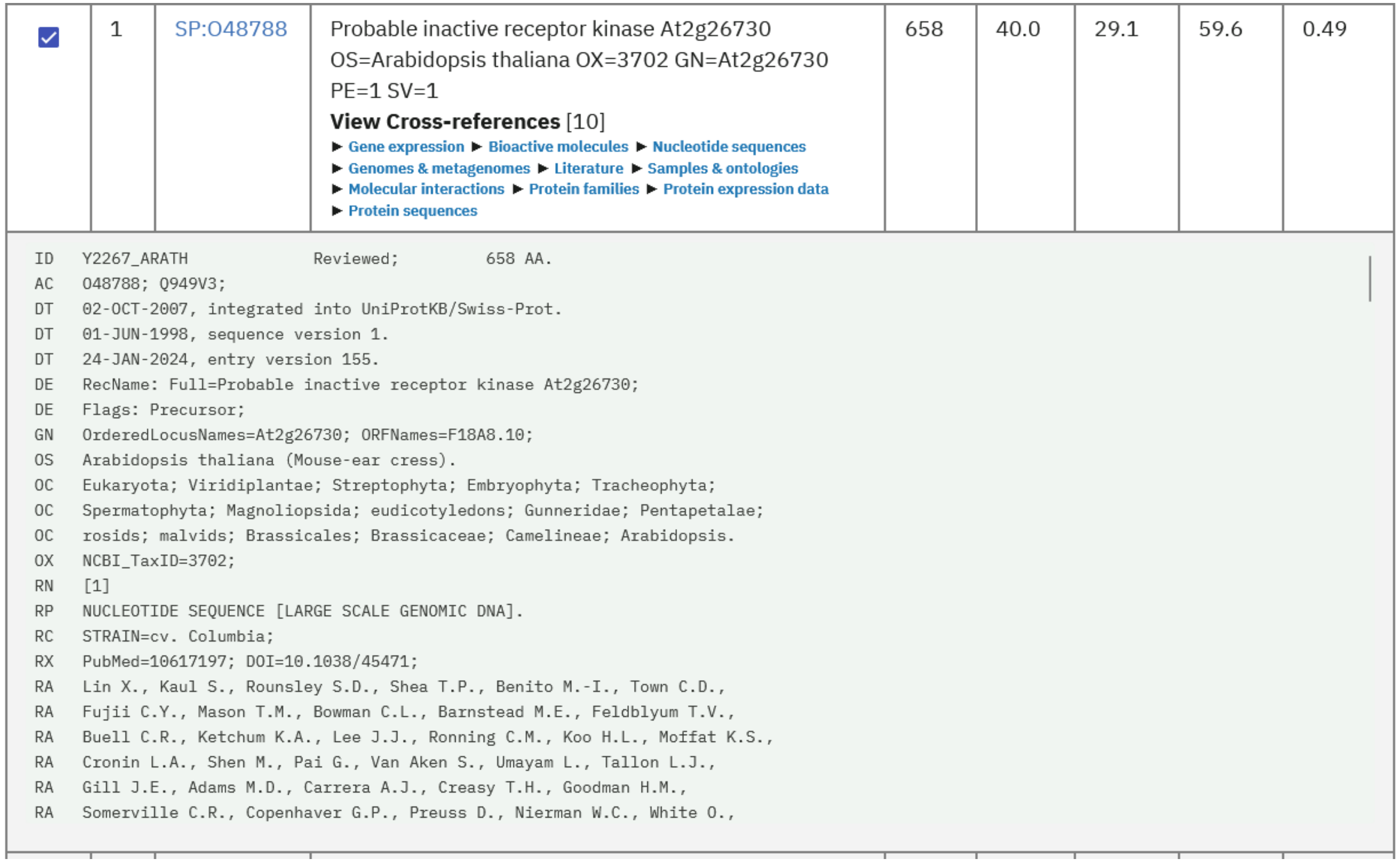
- You can download all found sequences in form of a single fasta format by clicking on “Download as FASTA” in the sidebar



FastA XXVIII – Web service – Result page VIII

- You‘re able to show UniProt annotations about the found sequences by clicking Show underneath Annotations in the sidebar (red rectangle)



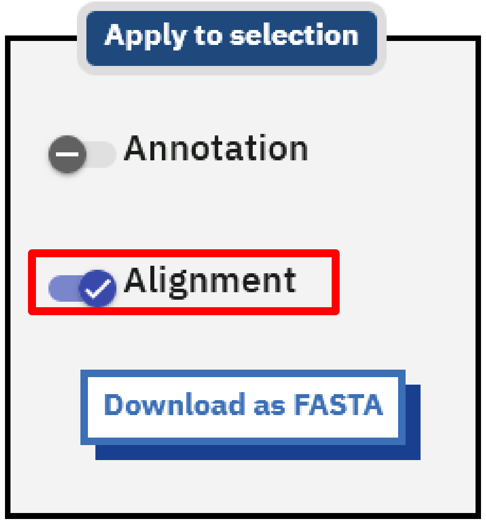


☑	1	SP:O48788	Probable inactive receptor kinase At2g26730 OS=Arabidopsis thaliana OX=3702 GN=At2g26730 PE=1 SV=1 **View Cross-references** [10] ▶ Gene expression ▶ Bioactive molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Literature ▶ Samples & ontologies ▶ Molecular interactions ▶ Protein families ▶ Protein expression data ▶ Protein sequences	658	40.0	29.1	59.6	0.49

```
ID   Y2267_ARATH             Reviewed;         658 AA.
AC   O48788; Q949V3;
DT   02-OCT-2007, integrated into UniProtKB/Swiss-Prot.
DT   01-JUN-1998, sequence version 1.
DT   24-JAN-2024, entry version 155.
DE   RecName: Full=Probable inactive receptor kinase At2g26730;
DE   Flags: Precursor;
GN   OrderedLocusNames=At2g26730; ORFNames=F18A8.10;
OS   Arabidopsis thaliana (Mouse-ear cress).
OC   Eukaryota; Viridiplantae; Streptophyta; Embryophyta; Tracheophyta;
OC   Spermatophyta; Magnoliopsida; eudicotyledons; Gunneridae; Pentapetalae;
OC   rosids; malvids; Brassicales; Brassicaceae; Camelineae; Arabidopsis.
OX   NCBI_TaxID=3702;
RN   [1]
RP   NUCLEOTIDE SEQUENCE [LARGE SCALE GENOMIC DNA].
RC   STRAIN=cv. Columbia;
RX   PubMed=10617197; DOI=10.1038/45471;
RA   Lin X., Kaul S., Rounsley S.D., Shea T.P., Benito M.-I., Town C.D.,
RA   Fujii C.Y., Mason T.M., Bowman C.L., Barnstead M.E., Feldblyum T.V.,
RA   Buell C.R., Ketchum K.A., Lee J.J., Ronning C.M., Koo H.L., Moffat K.S.,
RA   Cronin L.A., Shen M., Pai G., Van Aken S., Umayam L., Tallon L.J.,
RA   Gill J.E., Adams M.D., Carrera A.J., Creasy T.H., Goodman H.M.,
RA   Somerville C.R., Copenhaver G.P., Preuss D., Nierman W.C., White O.,
```

FastA XXIX – Web service – Result page IX

- Similar you can show the different alignments for your results by clicking on Show beneath Alignments in the sidebar (red rectangle)



☑	1	SP:O48788	Probable inactive receptor kinase At2g26730 OS=Arabidopsis thaliana OX=3702 GN=At2g26730 PE=1 SV=1 **View Cross-references** [10] ▶ Gene expression ▶ Bioactive molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Literature ▶ Samples & ontologies ▶ Molecular interactions ▶ Protein families ▶ Protein expression data ▶ Protein sequences	658	40.0	29.1	59.6	0.49

```
>SP:O48788 Y2267_ARATH Probable inactive receptor kinase At2g26730 OS=Arabidopsis
thaliana OX=3702 GN=At2g26730 PE=1 SV=1 (658 aa)
 initn:  52 init1:  52 opt: 134  Z-score: 160.8  bits: 40.0 E(1304960): 0.49
Smith-Waterman score: 134; 29.1% identity (59.6% similar) in 141 aa overlap (307-435:82-216)

        270       280       290       300       310         320       330       340
tr|B3J MRIGVMRDNSEGPWWLIWNFVSKEYADNYVPEDKPDPEPALPDGWEDMVSEV-VTSKIKWTMSADVPFDWANLDG----S
                                               .:.:    ..:. : :  .  .:...: :..::
SP:O48 ENRLQWNESDSACNWVGVECNSNQSSIHSLRLPGTGLVGQIPSGSLGRLTELRVLSLRSNRLSGQIPSDFSNLTHLRSLY
              50        60        70        80         90       100       110       120

             350       360       370       380       390       400       410       420
tr|B3J LMNNFTAGNYPDWATPVSGLDQLSMTLDSKTMTYEFAMPDGTTASGTYTLDEKGIYTFSGNVPSYHIGGGDIMFGADGNN
       :..:  .:..:   : ...: .:... .. : .  :.. . :  .: . : ..:   ::::.::  .:  :  :....::
SP:O48 LQHNEFSGEFPTSFTQLNNLIRLDISSNNFTGSIPFSVNNLTHLTGLF-LGNNG---FSGNLPSISLGLVD--FNVSNNN
             130       140       150       160         170          180         190

                    430       440       450       460       470       480       490
tr|B3J -------QLRILSIESAAGNVMGMWVGARSAEKDEYQAYHFIPNAGGGTSQPEATSIAVDNSKLVWGHLETDKNNFRIEL
              .:  .: :: .:::
SP:O48 LNGSIPSSLSRFSAESFTGNVDLCGGPLKPCKSFFVSPSPSPSLINPSNRLSSKKSKLSKAAIVAIIVASALVALLLLAL
         200       210       220       230       240       250       260       270
```

Examples B4DTL8 (good)

Examples B3JI28 (bad)

Exercise 01: FastA

- You have different fasta files containing one sequence each
- For at least two sequences run FastA searches using the parameters shown in the table on the right
- Are their any differences between the Search Runs?

Run	Parameters	Results
Run 1	Default	50
Run 2	BLOSUM62	50
Run 3	BLOSUM80	50
Run 4	PAM250	50
Run 5	PAM120	50
Run 6	MDM20	50
Run 7	VTML80	50
Run 8	BLOSUM50 but no gap penalties	50
Run 9	BLOSUM50 but highest gap penalties	50
Run 10	Default	250
Run 11	Default	1000

BLAST

BLAST I [10]

- **B**asic **L**ocal **A**lignment **S**earch **T**ool
- Idea: Only words that probably didn't evolve randomly are considered
- The "Matchscore" from a substitution matrix of the word or k-mer is his T-Score
- The T-Score has to be higher than a threshold T to be significant and considered further
  - That means instead of running a search will all k-mers from our query sequence we run the search only with specific k-mers
  - For example a k-mer like AAA would NOT be used by BLAST but by FastA
- The rest of the search is similar to FastA

BLAST II – Library Generation I [10]

- BLAST uses a library of k-mers consisting of k-mers from the query sequence that have a high enough T-score as well as k-mers similar to these ones
- First the query sequence is scanned for k-mers with a high enough T-score, in most cases T is set to 11 or 13 [11] depending on the purpose
- Afterwards all potential „neighbors“, which are in most cases k-mers which are very similar (difference of 1 letter in 3-mer) to the ones from the query sequence, are also added to the library

BLAST III – Library Generation II [10]

Test Sequence L N K C K T P Q G Q R L V N Q

P Q G 18 → k-mer

P E G 15
P R G 14
P K G 14
P N G 13
P D G 13
P M G 13

neighbors

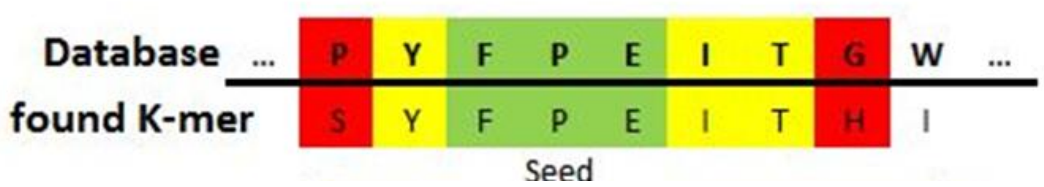
beneath threshold (T=13)

P Q A 12
P Q N 12
etc.

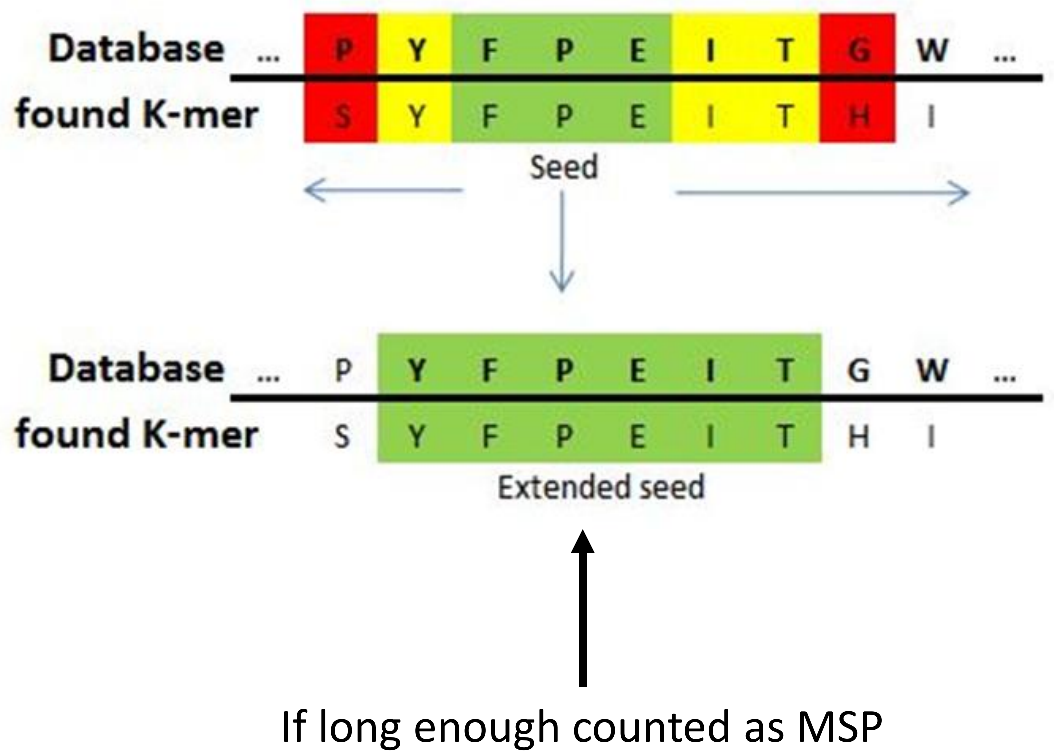
- On the left side the library generation for T=13 is shown for the example k-mer of PQG
- As we can see alongside the k-mer of PQG six other potential k-mers are included in the library
  - These are PEG, PRG, PKG, PNG, PDG and PMG

BLAST IV – Seeding I [10]

- The k-mers from the library are matched exactly to the sequences of the database (see image below)
- This process is called seeding and the alignment of k-mer and query sequence is called the seed
- This seed is than elongated in both directions along the database sequence by searching for additional matches/mismatches that have a positive score in the substitution matrix (next slide)



BLAST V – Seeding II [10]

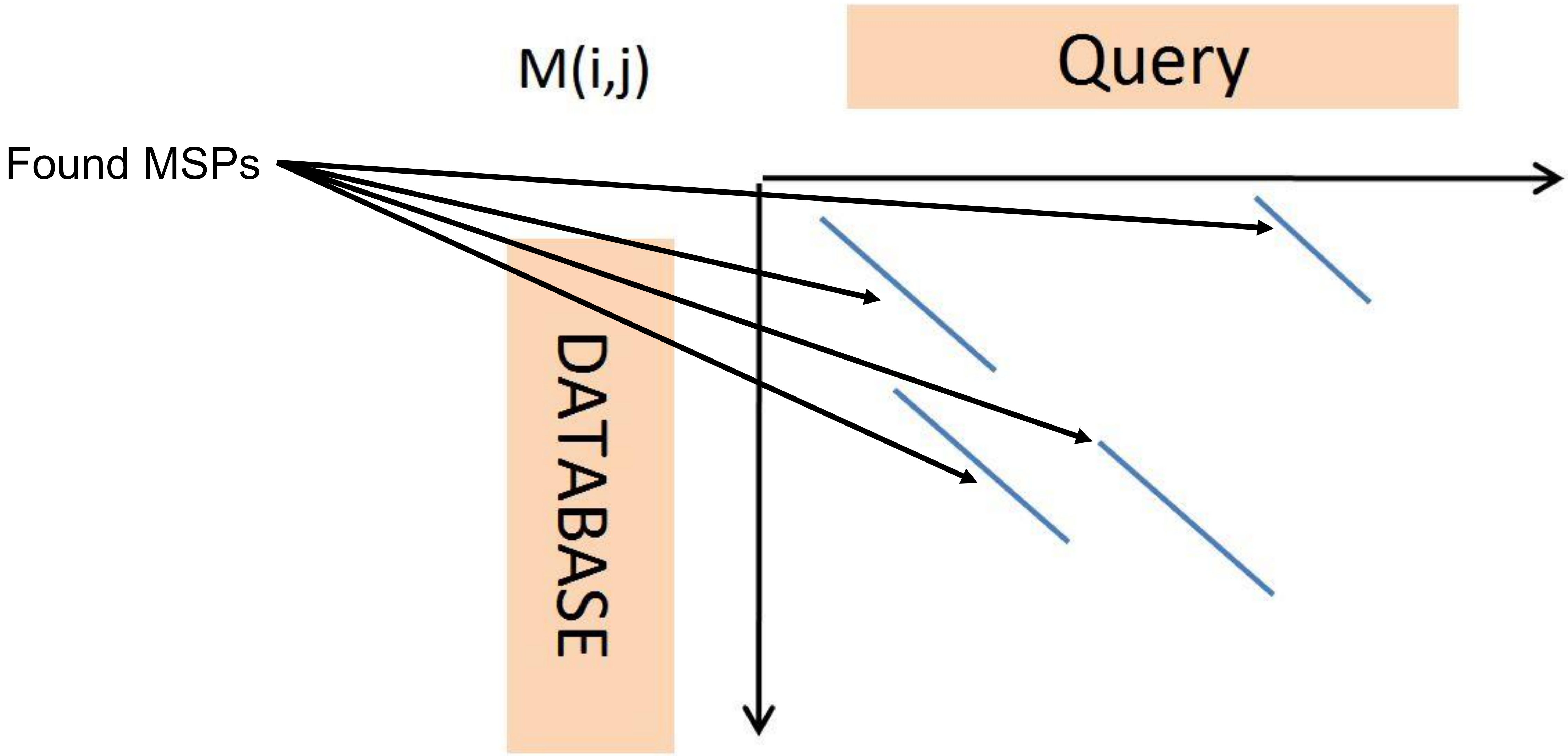


- If the extended seed is big enough it's counted as Maximal Segment Pair (MSP) and the sequence is taken over to the next step

BLAST VI – Chaining I [10]

- In the next step the algorithm tries to combine multiple MSPs to create High Scoring Pairs (HSPs)
- This is done using a banded Alignment algorithm to create an approximate Algorithm similar to the one FastA created
- Only HSPs that score high enough are considered as Hits and are handed over to the user after an optimal Alignment was generated

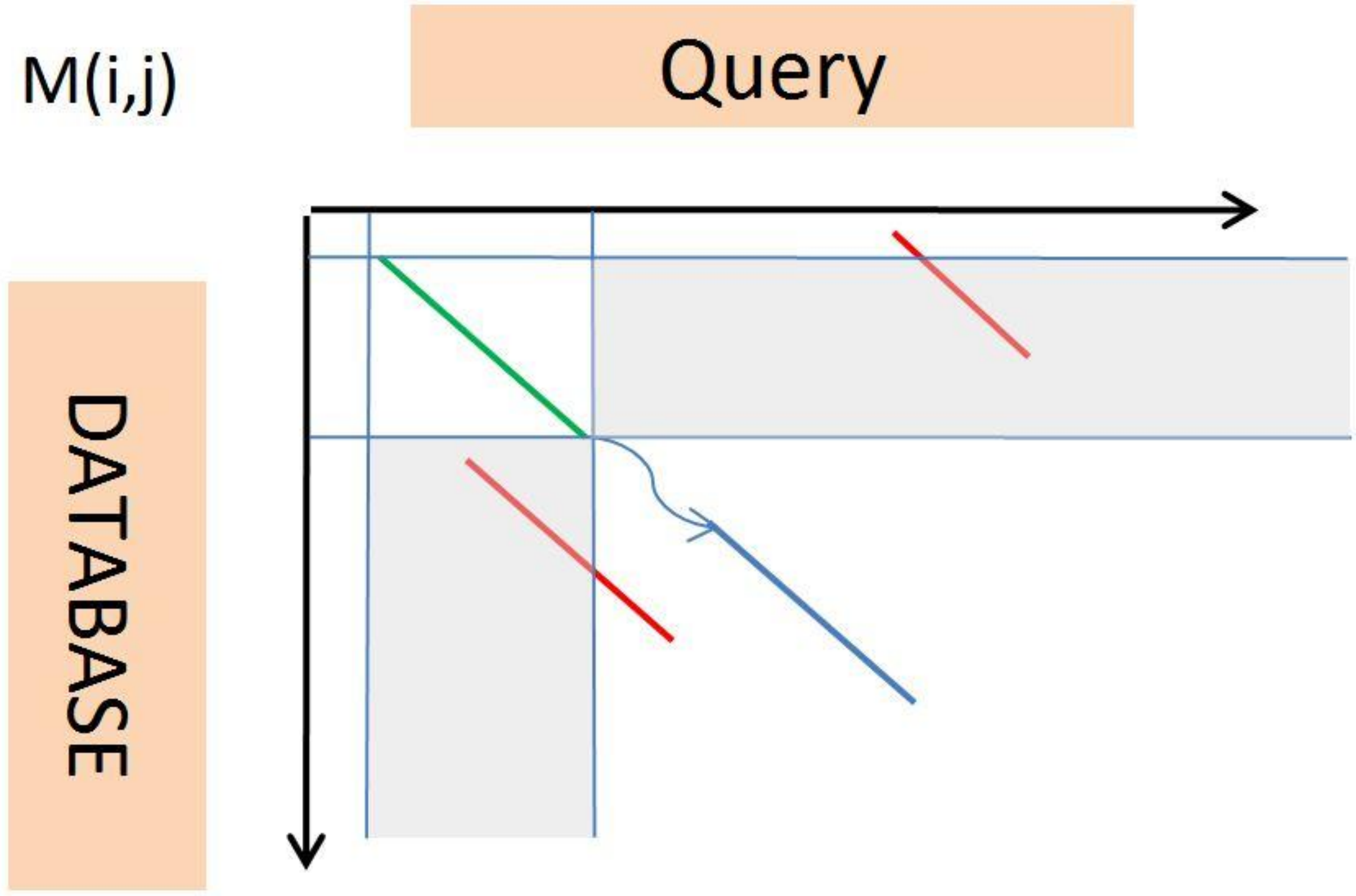
BLAST VII – Chaining II [10]



BLAST VIII – Chaining III [10]

- We pick the MSP that is closest to the top first (green) and ignore all other MSPs that have an overlap with it (red)

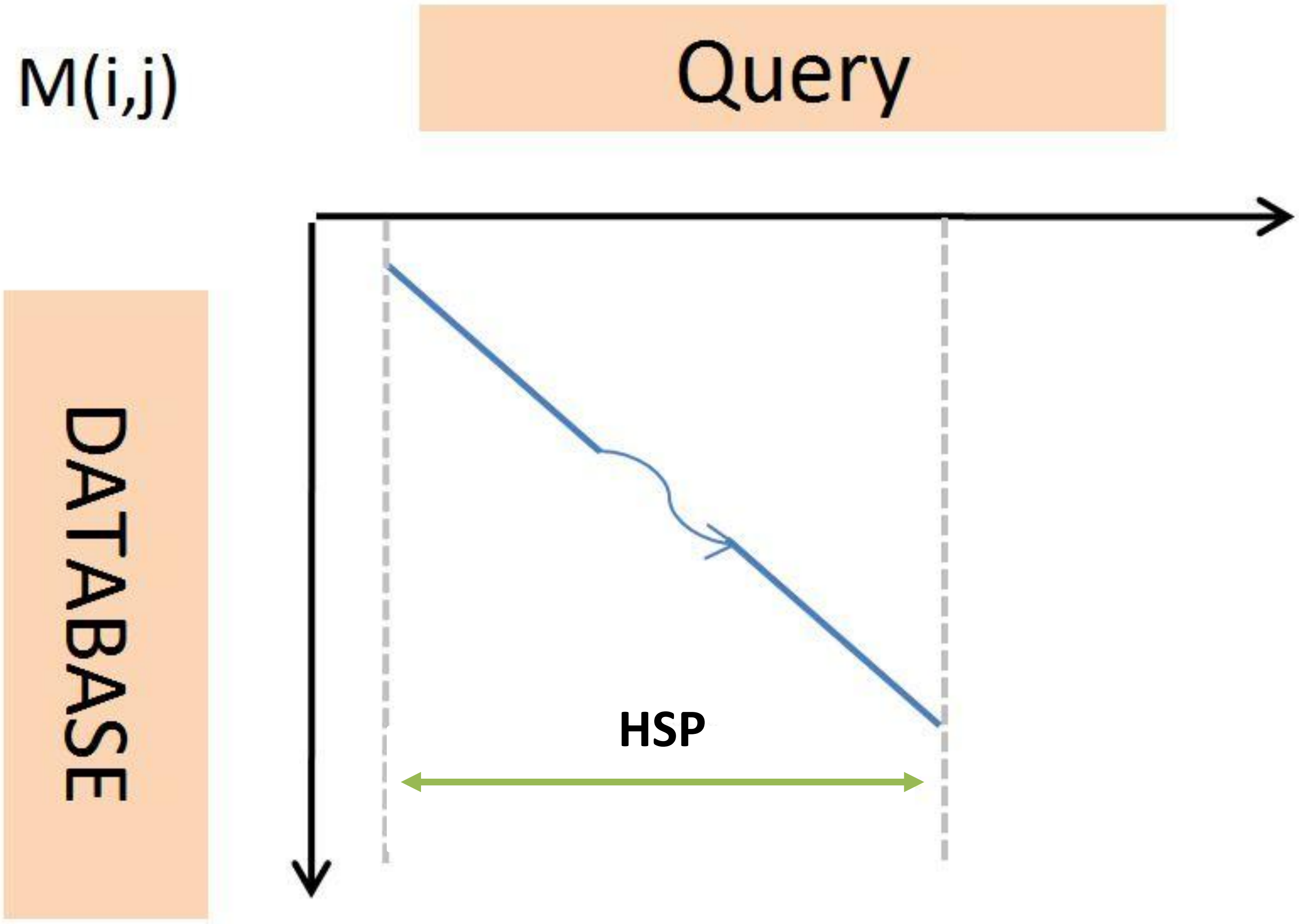
- Afterwards we pick the next nearest MSP to chain through the matrix



BLAST IX – Chaining IV [10]

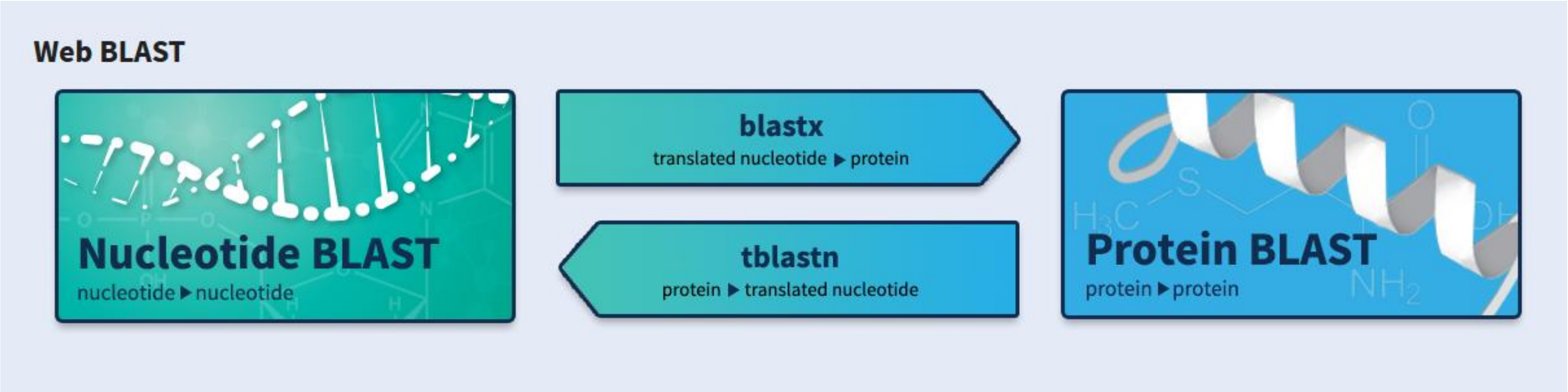
Once the chaining is completed we get a HSP, a so called High Scoring Pair

If this HSP scores above a certain threshhold the database sequence is seen as a hit and an optimal Alignment is calculated



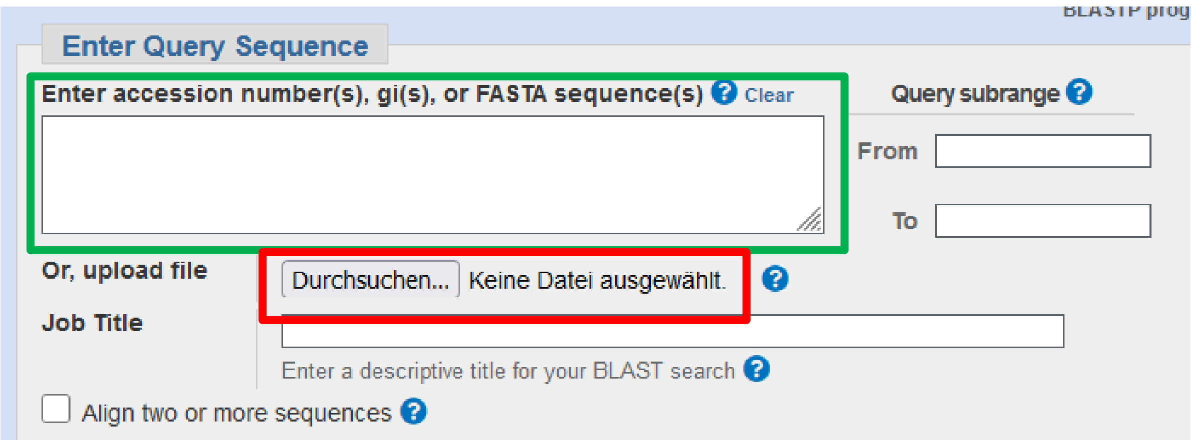
BLAST X – Web service

- Just like FastA we can also use a web service for working with BLAST
- It‘s available under the following links:
  - https://blast.ncbi.nlm.nih.gov/Blast.cgi
- Here we will focus on the NCBI one which leads us to a webpage that contains the different BLAST versions shown below



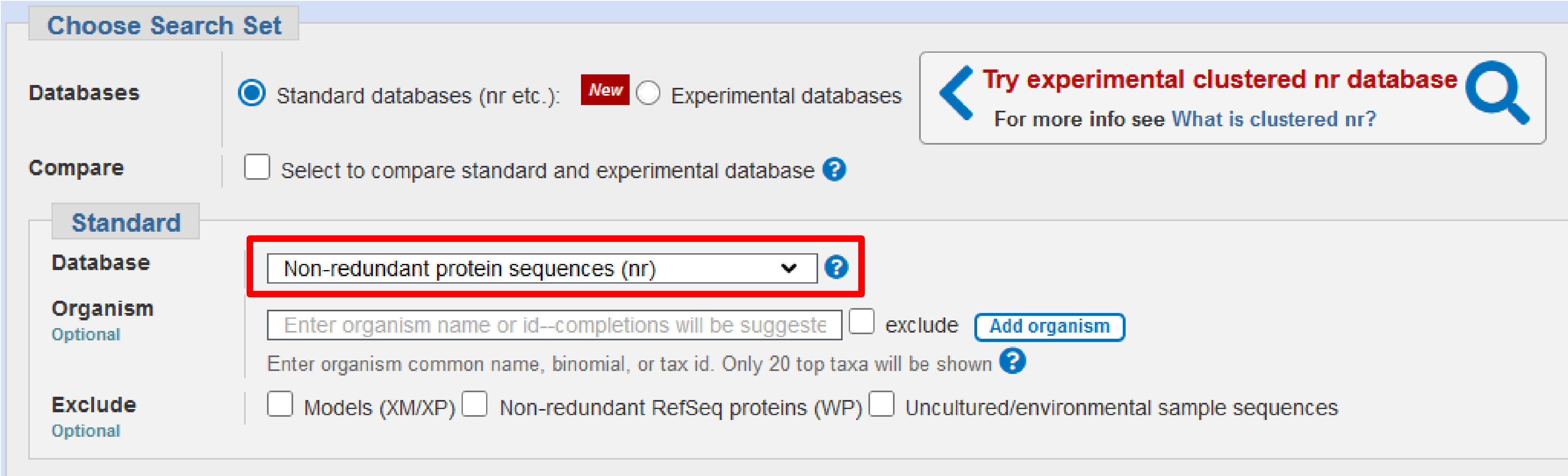
BLAST XI – Web service – Start Page I

- Once you clicked on Protein BLAST you are sent to the start page of the Protein BLAST
- At the top you can enter your sequence, either in the entry field (green rectangle) or you can upload a fasta file (red rectangle)
- BLAST allows you to use multiple sequences and runs a search for each of them separately



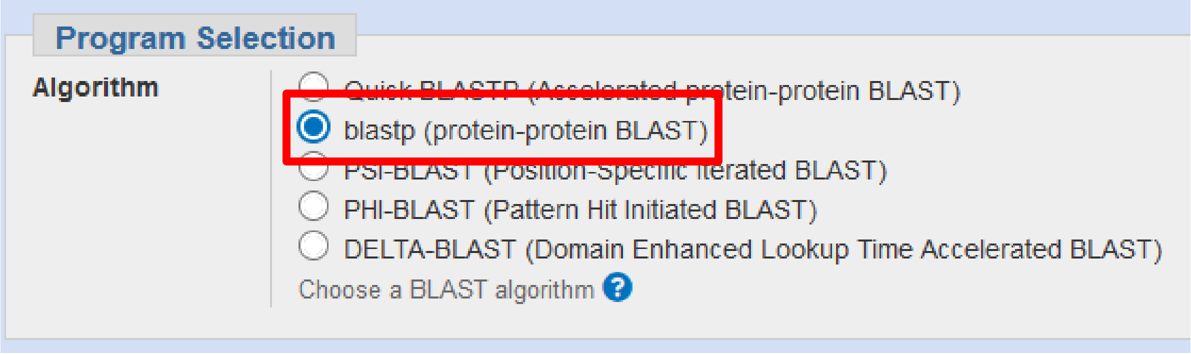
BLAST XII – Web service – Start Page II

- After you entered your sequences you can choose the database you want to search in (red rectangle)
- Per default the Non-redundant protein sequence database of the NCBI is selected
- You can also choose the UniProtKB/SwissProt



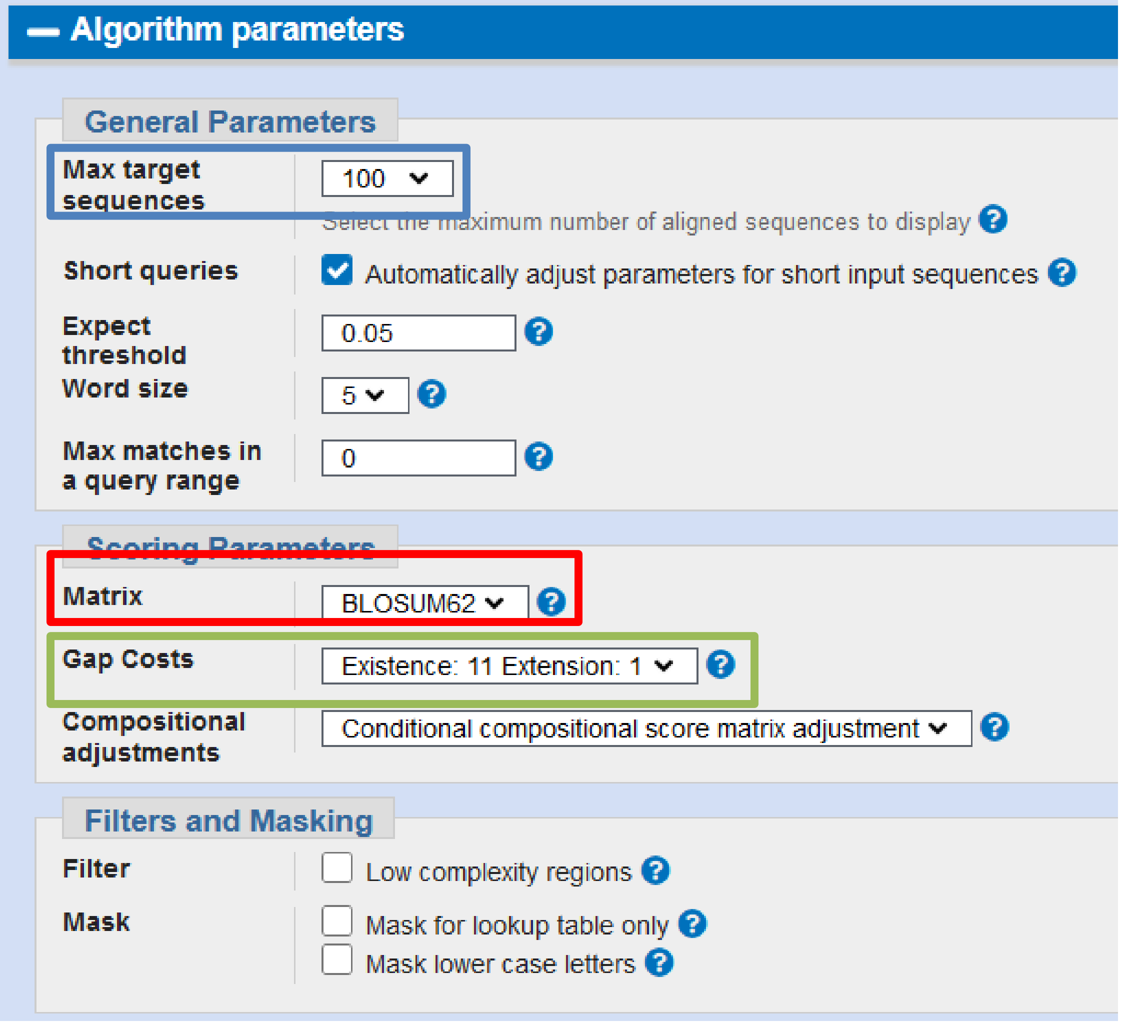
BLAST XIII – Web service – Start Page III

- Now you can select which type of Protein BLAST you want to execute
- For now we will only use the normal blastp (red rectangle)
- We will discuss the other three variants later on



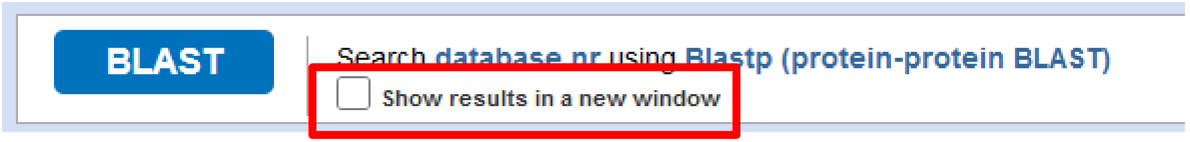
BLAST XIV – Web service – Start Page IV

- At the bottom of the page you can find a Button to adjust some parameters of your search (top right)
- If you click on it a Menu opens (bottom right) and here you can adjust for example the substitution matrix used (red) or the gap costs (green)
- You can also define how many results you want (blue)



BLAST XV – Web service – Start Page V

- Once everything is set up to your liking you can start your search by clicking on BLAST
- If you want to start multiple searches you can activate the check button “Show results in a new window”
- By doing so the result window will appear in a new Tab in your Browser and you can upload a new sequence to start a different search



BLAST XVI – Web service – Results I

- Once your search is finished the result page will be shown and at the top you will see some information about your search
- It will look differently whether you used one sequence (right) or multiple sequences (left), in the latter case you can swith between different results (red)

Job Title	**tr|B4DTL8|B4DTL8_HUMAN cDNA FLJ61389, highly**
RID	6T0G1H2P01N *Search expires on 05-24 17:06 pm* Download All
Results for	1:lcl|Query_343729 tr|B4DTL8|B4DTL8_HUMAN cDNA FLJ61389, hi
Program	BLASTP Citation
Database	nr See details
Query ID	lcl|Query_343729
Description	tr|B4DTL8|B4DTL8_HUMAN cDNA FLJ61389, highly simil ...
Molecule type	amino acid
Query Length	824
Other reports	Distance tree of results Multiple alignment MSA viewer

Job Title	**tr|B3JI28|B3JI28_9BACT Uncharacterized**
RID	6T1RTVTU016 *Search expires on 05-24 17:27 pm* Download All
Program	BLASTP Citation
Database	nr See details
Query ID	lcl|Query_183812
Description	tr|B3JI28|B3JI28_9BACT Uncharacterized protein OS=Ph ...
Molecule type	amino acid
Query Length	608
Other reports	Distance tree of results Multiple alignment MSA viewer

BLAST XVII – Web service – Results II

- The main part of the results page is the tabular overview of the found sequences
- They are listed with different scores (red), the E-value (green) and the percentage identity of the Alignment (blue)
- If one sequence produced multiple Alignments Total Score would be the sum of all scores while Max Score would show the highest single score reached

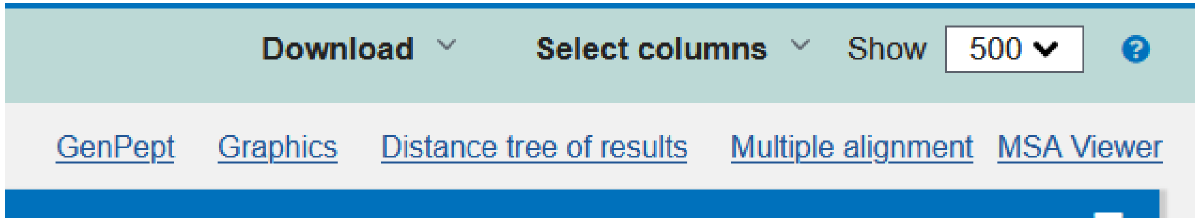
Sequences with E-value BETTER than threshold

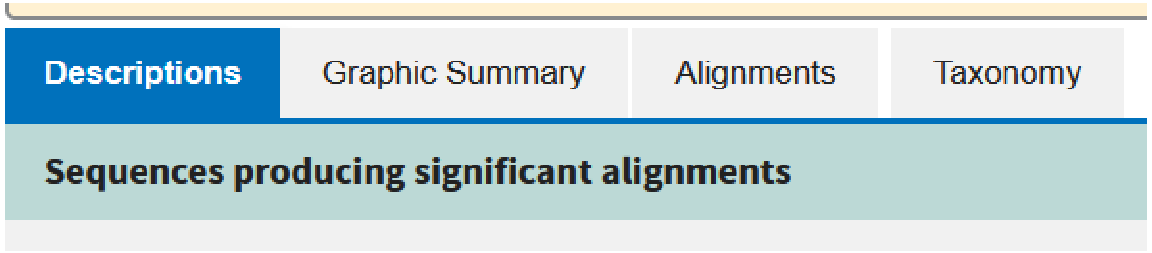
☑ select all *500 sequences selected* **PSI-BLAST iteration 1**

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	Select for PSI blast	Used to build PSSM	Newly added
☑	hypothetical protein BACCOP_01540 [Phocaeicola coprocola DSM 17136]	Phocaeicola coprocol...	1252	1252	100%	0.0	100.00%	608	EDV01452.1	☑		
☑	TPA: hypothetical protein [Bacteroides sp.]	Bacteroides sp.	1249	1249	100%	0.0	99.67%	608	HCM10933.1	☑		
☑	hypothetical protein [Phocaeicola coprocola]	Phocaeicola coprocola	1235	1235	98%	0.0	100.00%	600	WP_040312043.1	☑		
☑	hypothetical protein [Phocaeicola coprocola]	Phocaeicola coprocola	1233	1233	98%	0.0	99.83%	600	WP_278964951.1	☑		
☑	hypothetical protein [Bacteroides sp.]	Bacteroides sp.	1232	1232	98%	0.0	99.67%	600	MBS4813119.1	☑		
☑	TPA: hypothetical protein [Phocaeicola coprocola]	Phocaeicola coprocola	1232	1232	98%	0.0	99.67%	600	HJF08188.1	☑		
☑	hypothetical protein [Phocaeicola coprocola]	Phocaeicola coprocola	1230	1230	98%	0.0	99.50%	600	WP_118484359.1	☑		

BLAST XVIII – Web service – Results III

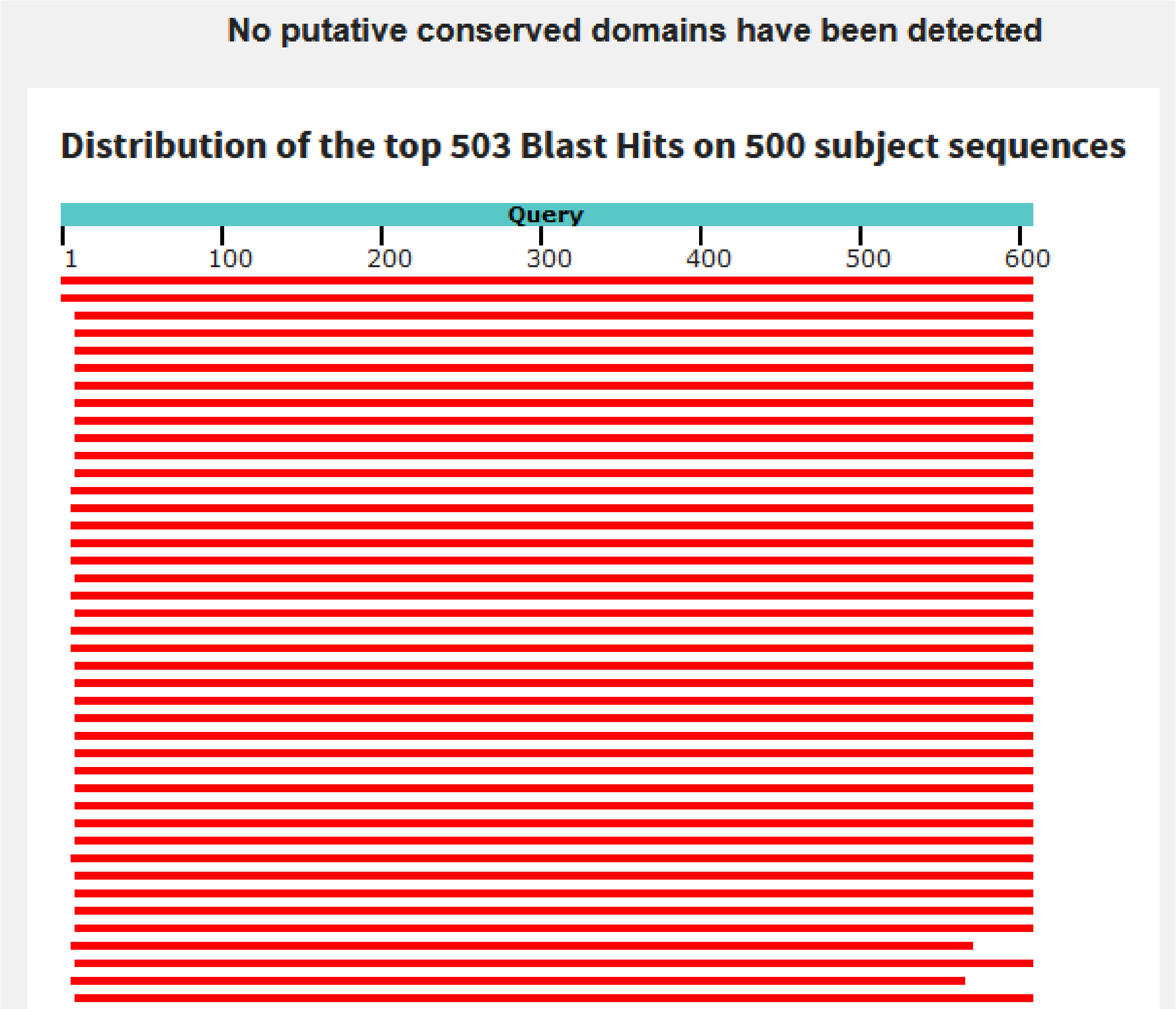
- At the top of the result table you can find some tabs that allow you to switch between different ways of displaying the results (right)
- You can also Download the found sequences in different formats as well as calculate a MSA using the found sequences by using the options on the top of your result table (left)





BLAST XIX – Web service – Results IV

- By clicking on "Graphic Summary" you're able to get a graphical overview of the Alignments for all sequences your search found
- At the top your query sequence is shown schematically and below it you can see the database sequences as red bars
- The position of these bars reflect the Alignment between the sequences



BLAST XX – Web service – Results V

- Under “Alignment” you can look up the calculated optimal Alignments between your query sequence (always at the top) and the different found Hits of the database (always at the bottom)
- You can also see the identity (red), how many positive match/mismatch scores you have (green) and the number of gaps (blue)

Download GenPept Graphics

hypothetical protein BACCOP_01540 [Phocaeicola coprocola DSM 17136]

Sequence ID: EDV01452.1 Length: 608 Number of Matches: 1

Range 1: 1 to 608 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps
1252 bits(3240)	0.0	Compositional matrix adjust.	608/608(100%)	608/608(100%)	0/608(0%)

```
Query  1    MELKTQEVMKKLISKLYTAWLIITVGLFSACTPDSFELEGKDVTVDDLVEGIAFSITHDS  60
            MELKTQEVMKKLISKLYTAWLIITVGLFSACTPDSFELEGKDVTVDDLVEGIAFSITHDS
Sbjct  1    MELKTQEVMKKLISKLYTAWLIITVGLFSACTPDSFELEGKDVTVDDLVEGIAFSITHDS  60

Query  61   ENPNIVYLKSLMPSSYQVCWQHPQGRSQEREVTLQMPFEGKYEVTFGVQTRGGIVYGNPA  120
            ENPNIVYLKSLMPSSYQVCWQHPQGRSQEREVTLQMPFEGKYEVTFGVQTRGGIVYGNPA
Sbjct  61   ENPNIVYLKSLMPSSYQVCWQHPQGRSQEREVTLQMPFEGKYEVTFGVQTRGGIVYGNPA  120

Query  121  TFTIDSFCADFVNDDLWTYLTGGVNNEKVWIFDNGSYGYAAGEMTYADPSTTVEWNNWSA  180
            TFTIDSFCADFVNDDLWTYLTGGVNNEKVWIFDNGSYGYAAGEMTYADPSTTVEWNNWSA
Sbjct  121  TFTIDSFCADFVNDDLWTYLTGGVNNEKVWIFDNGSYGYAAGEMTYADPSTTVEWNNWSA  180

Query  181  NWDPGVGHTGDDAIWESTMTFGLKGGANVTVYNSSSKETASGTFMLNTSNHTITFTDCEL  240
            NWDPGVGHTGDDAIWESTMTFGLKGGANVTVYNSSSKETASGTFMLNTSNHTITFTDCEL
Sbjct  181  NWDPGVGHTGDDAIWESTMTFGLKGGANVTVYNSSSKETASGTFMLNTSNHTITFTDCEL  240

Query  241  LHTPSWSDRSTNWSRDLKLLELDENHMRIGVMRDNSEGPWWLIWNFVSKEYADNYVPEDK  300
            LHTPSWSDRSTNWSRDLKLLELDENHMRIGVMRDNSEGPWWLIWNFVSKEYADNYVPEDK
Sbjct  241  LHTPSWSDRSTNWSRDLKLLELDENHMRIGVMRDNSEGPWWLIWNFVSKEYADNYVPEDK  300

Query  301  PDPEPALPDGWEDMVSEVVTSKIKWTMSADVPFDWANLDGSLMNNFTAGNYPDWATPVSG  360
            PDPEPALPDGWEDMVSEVVTSKIKWTMSADVPFDWANLDGSLMNNFTAGNYPDWATPVSG
Sbjct  301  PDPEPALPDGWEDMVSEVVTSKIKWTMSADVPFDWANLDGSLMNNFTAGNYPDWATPVSG  360

Query  361  LDQLSMTLDSKTMTYEFAMPDGTTASGTYTLDEKGIYTFSGNVPSYHIGGGDIMFGADGN  420
            LDQLSMTLDSKTMTYEFAMPDGTTASGTYTLDEKGIYTFSGNVPSYHIGGGDIMFGADGN
Sbjct  361  LDQLSMTLDSKTMTYEFAMPDGTTASGTYTLDEKGIYTFSGNVPSYHIGGGDIMFGADGN  420

Query  421  NQLRILSIESAAGNVMGMWVGARSAEKDEYQAYHFIPNAGGGTSQPEATSIAVDNSKLVW  480
            NQLRILSIESAAGNVMGMWVGARSAEKDEYQAYHFIPNAGGGTSQPEATSIAVDNSKLVW
Sbjct  421  NQLRILSIESAAGNVMGMWVGARSAEKDEYQAYHFIPNAGGGTSQPEATSIAVDNSKLVW  480

Query  481  GHLETDKNNFRIELYNMYGETASASPIDPASIVFDYSMELTFTISGLTGDAATKEYNAGL  540
            GHLETDKNNFRIELYNMYGETASASPIDPASIVFDYSMELTFTISGLTGDAATKEYNAGL
Sbjct  481  GHLETDKNNFRIELYNMYGETASASPIDPASIVFDYSMELTFTISGLTGDAATKEYNAGL  540

Query  541  MCTADGWWPAYSGTSDVKVKGDGTYTIKMKPEAAYNGVIVFVIDIFDMFSDIADLDAVNV  600
            MCTADGWWPAYSGTSDVKVKGDGTYTIKMKPEAAYNGVIVFVIDIFDMFSDIADLDAVNV
Sbjct  541  MCTADGWWPAYSGTSDVKVKGDGTYTIKMKPEAAYNGVIVFVIDIFDMFSDIADLDAVNV  600

Query  601  TIDSLKIL  608
            TIDSLKIL
Sbjct  601  TIDSLKIL  608
```

Examples B4DTL8 (good)

Examples B3JI28 (bad)

Exercise 02: WNT proteins

- Run a BLAST Search with the fasta-file P41221.fasta against the SwissProt and the RefSeq Protein
- Please include only results from the organism homo sapiens
- How many results can you get from these two searches?
- Can you find the sequence with the UniProt identifier P56704 in the results for the search against the SwissProt?
- Afterwards download all results in a single fasta-file and create at least two multiple sequence alignments using Clustal Omega, MUSCLE, MAFFT, T-COFFEE or K-Align of the two result sets
- What would you say about the quality of the MSAs?

Exercise 03: Structure Search

- Use the fasta-files OIP98664.fasta, WP_223908953.fasta and NP_477249.fasta to run protein BLASTS against the pdb database with default settings
- How many results does each of the three searches produce?
- Repeat the searches but this time use a wordsize of 2
- Are there any differences in the results?
- Try to find sequences in the pdb for the query ranges 1-250, 1-300 and 1-350 for NP_477249.fasta to see whether you can find structures for this part
- How do the results of these searches look like?
- Go to the UniProt and search for the sequence using the identifier of it
- What kind of structures can you find there?

Exercise 04: Similar structures

- Use the fasta file rcsb_pdb_1BGY.fasta and run a normal blastp search against the pdb database
- How many results can you find for the different sequences present in the file?
- How many of these results do you think are useful?
- Can you find the structure itself in the results?

Exercise 05: TRUD Domains and BLAST

- Use the fasta files M3Z3I3.fasta, A0A5N3XQ52.fasta and A0A9D3NLI7.fasta to run protein Blasts against the swissprot, the RefSeq Select, the RefSeq and the nonredundant protein database
- How many results can you find for each search?
- Look at the results for RefSeq and the nonredundant database in detail and determine how many results you can find where the existence of the protein is known
  - Tip: the name shouldn’t contain the word hypothetical and the Accessions shouldn’t start with XP

Exercise 06: Protein BLAST

- Run BLAST searches using the same sequences as in Exercise 1
- Start multiple Runs using the different search parameters shown in the table on the right

Run	**Parameters**	**Results**
Run 1	Default	100
Run 2	BLOSUM50	100
Run 3	BLOSUM90	100
Run 4	PAM250	100
Run 5	PAM30	100
Run 6	PAM70	100
Run 7	BLOSUM62 & word size 2	100
Run 8	BLOSUM62 and minimal gap penalties	100
Run 9	Default	250
Run 10	Default	1000

BLAST Variants

BLAST Variants I

- Until now we talked about protein BLAST but there are additional versions of this algorithm as well
- These variants work with Nucleotide sequences or translated nucleotide sequences to search for proteins
- These are called:
  - Blastn
  - Blastx
  - tBlastn
  - tBlastx

BLAST Variants II

- When working with nucleotide sequences in BLAST searches the overall time our algorithm needs for one search increases dramatically
- A normal nucleotide BLAST already takes around 10 to 20 times of a Protein BLAST
- If we use the translational methods like blastx we can assume that it will take at least 100 times as long as a simple Protein BLAST

BLAST Variants III

- Therefore it could happen especially when using tblastx and tblastn that you won‘t get any results and instead the message shown below appears
- This message means your search took too long for the server and you were kicked out
- If you come across such a message in your later career you should discuss using a local installation of the blast+ suite instead of the webserver

> There was a problem with the search. Please, contact Help Desk and include RID 6W460D6Y013.
> CPU usage limit was exceeded. You may need to change your search strategy. Helpful changes include reducing the number of queries, choosing a smaller database and setting an organism limit. You may also need to adjust the Algorithm parameters in the bottom section of the form. Choose a smaller number of target sequences, set a smaller Expect cut-off, use a larger word-size and turn on species specific repeats.

BLAST Variants IV – blastn

- Is the Nucleotide Blast and allows us to search with nucleotide sequences in a database of nucleotide sequences
- Returns nucleotide sequences that are similiar to the search query
- In general the algorithm follows the same principles we saw when talking about the Protein BLAST
- However some parameter adjustments are needed:
  - Instead of using k-mers of 2 we use k-mers of at least 16, with a default of 28
  - Instead of using a scoring matrix, we have fixed Match (1) and Mismatch scores (-2)
  - Instead of Gap-open and Gap-extension costs we use linear gap costs of 0 for megablast and -5 for other blastn versions

BLAST Variants V – blastx I

- Searches the protein databases with the help of a translated nucleotide sequence
- Uses a nucleotide sequence as input and translates it into six different protein sequences
- It than runs a Protein BLAST search for each of these translated sequences
- Returns protein sequences as outputs of the different Protein BLAST searches
- We get back the best overall results from all six searches, that means if one search produces only “bad results” it could not be part of the overall results

BLAST Variants VI – Reading Frames [11]

ACGTACGTACGTACGTACGT

TGCATGCATGCATGCATGCA

Frame +1 ACGTACGTACGTACGTACGT

Frame +2 ACGTACGTACGTACGTACGT

Frame +3 ACGTACGTACGTACGTACGT

Frame -1 TGCATGCATGCATGCATGCA

Frame -2 TGCATGCATGCATGCATGCA

Frame -3 TGCATGCATGCATGCATGCA

BLAST Variants VII – blastx II

- On the results page you can look up the Alignments of your query sequence with the translated protein sequences
- For each Alignment the corresponding reading Frame is indicated at the top right corner of the Alignment (red circle)

Download GenPept Graphics

MFS transporter [Escherichia coli]

Sequence ID: MBA1935651.1 Length: 113 Number of Matches: 1

Range 1: 50 to 99 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
58.2 bits(139)	6e-09	Compositional matrix adjust.	46/50(92%)	46/50(92%)	0/50(0%)	-1

```
Query  150   LTLL*GRETETSLVKMNlpgltllvlgvggLQIMLDKRRDLDWLNSSTII  1
             LTLL GRETETS VKMNLPGLTLLVLGVGGLQIMLDK RDLDW NSSTII
Sbjct  50    LTLLKGRETETSPVKMNLPGLTLLVLGVGGLQIMLDKGRDLDWFNSSTII  99
```

BLAST Variants VIII – tblastn I

- Searches the translated nucleotide database with the help of a protein query
- This is done by converting the nucleotide sequences into all possible six reading frames and thus running a BLAST search against six databases
- It may be used to map a protein to genomic DNA since you can determine the position of the corresponding region in the nucleotide sequence that was used for the corresponding Alignment

BLAST Variants IX – tblastn II

- The most information can be found when looking at the Alignments of your results in detail
- Not only can you again see the corresponding reading frame (red) but you can also see the position in the corresponding genome that was aligned (green circles)

Bacteroides xylanisolvens strain funn3 chromosome, complete genome

Sequence ID: CP045612.1 Length: 6570539 Number of Matches: 1

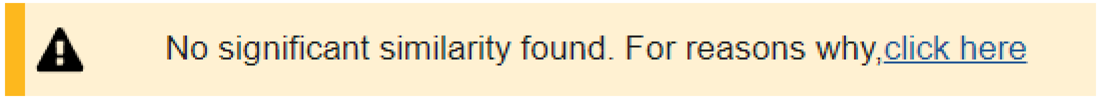
Range 1: 3487328 to 3489130 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
1047 bits(2708)	0.0	Compositional matrix adjust.	498/602(83%)	551/602(91%)	1/602(0%)	+2

```
Query  7        EVMKKLISKLYTAWLIITVGLFSACTPDSFELEGKDVTVDDLVEGIAFSITHDSENPNIV  66
                EVM K++SKLY A L   V L SACTP+++EL  +D+  DDL EGIAFSITHD+ NPNIV
Sbjct  3487328  EVMNKIVSKLYIALLATLVTLLSACTPETYELGSQDIVSDDLAEGIAFSITHDAVNPNIV  3487507

Query  67       YLKSLMPSSYQVCWQHPQGRSQEREVTLQMPFEGKYEVTFGVQTRGGIVYGNPATFTIDS  126
                YLKSLMPS+YQVCW+HPQGRSQ  EV LQMPFEG+Y V FGVQTR GIVYG  A FTIDS
Sbjct  3487508  YLKSLMPSNYQVCWEHPQGRSQNAEVKLQMPFEGEYSVKFGVQTRAGIVYGPAAKFTIDS  3487687

Query  127      FCADFVNDDLWTYLTGGVNNEKVWIFDNGSYGYAAGEMTYADPSTTVEWNNWSANWDPGV  186
                FCA+FVND LWTYL+GGVNNEKVWIFDNGSYG+AAGE+TYADPSTTVEWNNWSANWDPGV
Sbjct  3487688  FCAEFVNDALWTYLSGGVNNEKVWIFDNGSYGFAAGEVTYADPSTTVEWNNWSANWDPGV  3487867

Query  187      GHTGDDAIWESTMTFGLKGGANVTVYNSSSKETASGTFMLNTSNHTITFTDCELLHTPSW  246
                GH+GDDAIWESTMTFGL GGANVTVYNSSS+ TA GTFMLN  +HTITFTDCELLHTPSW
Sbjct  3487868  GHSGDDAIWESTMTFGLTGGANVTVYNSSSQSTAVGTFMLNVDDHTITFTDCELLHTPSW  3488047

Query  247      SDRSTNWSRDLKLLELDENHMRIGVMRDNSEGPWWLIWNFVSKEYADNYVPEDKPDPEPA  306
                SDRSTNW+R+LKLLELDENHMRIGVMRDNSEGPWWL+WNFVSKEYADNYVPED+PDPEPA
Sbjct  3488048  SDRSTNWNRNLKLLELDENHMRIGVMRDNSEGPWWLVWNFVSKEYADNYVPEDRPDPEPA  3488227

Query  307      LPDGWEDMVSEVVTSKIKWTMSADVPFDWANLDGSLMNNFTAGNYPDWATPVSGLDQLSM  366
                LPDGWEDMVSEVVT++IKWTMS DVPFDWANLDGSLMNNFTAGNYP+WATPVSGL++L++
Sbjct  3488228  LPDGWEDMVSEVVTTEIKWTMSPDVPFDWANLDGSLMNNFTAGNYPEWATPVSGLEKLAL  3488407

Query  367      TLDSKTMTYEFAMPDGTTASGTYTLDEKGIYTFSGNVPSYHIGGGDIMFGADGNNQLRIL  426
                TL+S TMTY   +PDGT++ GTY+LD+KGIYTFS  +P+ HIGGGDIMFGAD NNQLRIL
Sbjct  3488408  TLNSSTMTYSLVLPDGTSSDGTYSLDKKGIYTFSSALPACHIGGGDIMFGADANNQLRIL  3488587

Query  427      SIESAAGNVMGMWVGARSAEKDEYQAYHFIPNAGGGTSQPEATSIAVDNSKLVWGHLETD  486
                 IESA G+V+GMW+GARS+EKDEYQAYHF+PNA GG+S+PEAT+I VDN KLVWGHLE D
Sbjct  3488588  RIESAGGSVIGMWLGARSSEKDEYQAYHFVPNA-GGSSEPEATTITVDNHKLVWGHLEND  3488764

Query  487      KNNFRIELYNMYGETASASPIDPASIVFDYSMELTFTISGLTGDAATKEYNAGLMCTADG  546
                KNNFRIELYN YG+TASASP+DPASIVFDYSMELTFTISGL+GDAATKEYNAGLMCTA G
Sbjct  3488765  KNNFRIELYNQYGQTASASPVDPASIVFDYSMELTFTISGLSGDAATKEYNAGLMCTASG  3488944

Query  547      WWPAYSGTSDVKVKGDGTYTIKMKPEAAYNGVIVFVIDIFDMFSDIADLDAVNVTIDSLK  606
                WWP+YSGTSDVKVKG+GTYTI +KPEAAYNGVIVFVIDI DMFSDIA+ D VNVTID+LK
Sbjct  3488945  WWPSYSGTSDVKVKGNGTYTINIKPEAAYNGVIVFVIDIIDMFSDIAEPDKVNVTIDTLK  3489124

Query  607      IL  608
                IL
Sbjct  3489125  IL  3489130
```

BLAST Variants X – tblastx [12]

- Searches the translated nucleotide database with the help of a nucleotide query
- This is done by converting the nucleotide sequence into all possible six reading frames
- Compares these six reading frames with a database of six reading frames
- For discovering novel genes in the nucleotide sequences
- Is not recommended for large queries since it takes a lot of time and resources
- More often than not you will see the message shown below



blastn Example KR85597

tblastn Examples B4DTL8 (good)

tblastn Examples B3JI28 (bad)

blastx Example KR85597

Exercise 07: Hepatitis C NS5B

- Use the accession codes KR855587.1 and KR855596.1 to run normal nucleotide BLASTs against the “Nucleotide Collection” database while showing the best 500 Hits
- Afterwards use blastx with the same sequences against the “non-redundant protein sequences” database, against the “UniProtKb/Swiss-Prot” database and against the “Protein Data Bank proteins” database
- Are there any big differences between the results for the two sequences?
- Create pairwise Alignments using Needleman-Wunsch and Smith-Waterman to see how closely these two sequences are related to each other

Exercise 08: Jurassic Park

- The sequences for this exercise can be found in the file ABI_Exercises.pdf
- Use nucleotide-nucleotide BLAST against the default nucleotide database, nr, to identify the real source of the first sequence
- NCBI scientist Mark Boguski noticed this obvious "contaminant" and supplied Crichton with a better sequence, for the sequel, The Lost World
- Identify the most likely source of this sequence using nucleotide-nucleotide BLAST
- Mark imbedded his name in the sequence he provided
- To see Mark's name use the translating BLAST (blastx) page with the sequence below. (Look for MARK WAS HERE NIH).

Exercise 09: Translating TRUD Domains

- Use the fasta files M3Z3I3.fasta, A0A5N3XQ52.fasta and A0A9D3NLI7.fasta to run tblastn searches against the databases RefSeq_Select_RNA, RefSeq_RNA and Core_NT.
- How many results can you find for each run?
- How many of these results aren’t predicted?
- Can you find a result with 100% sequence identity?

Exercise 10: Blast Alternatives

- You have ten fasta files containing Nucleotide sequences
- Use at least two of them to run the searches shown in the table on the right
- Only change the listed parameter and leave the rest on default
- Use at least two protein sequences to run the searches 11 to 15 shown below with tblastn

Run	Program	Parameters	Alignments
Run 1	blastn	Default	100
Run 2	blastx	Default	100
Run 3	blastn	word size 16	100
Run 4	blastn	word size 64	100
Run 5	blastn	Match/Mismatch Score 1,-4	100
Run 6	blastn	Match/Mismatch Score 5,-4	100
Run 7	blastx	word size 2	100
Run 8	blastx	BLOSUM45	100
Run 9	blastx	PAM250	100
Run 10	blastx	PAM30	100
Run 11	tblastn	Default	100
Run 12	tblastn	word size 2	100
Run 13	tblastn	BLOSUM45	100
Run 14	tblastn	PAM250	100
Run 15	tblastn	PAM30	100

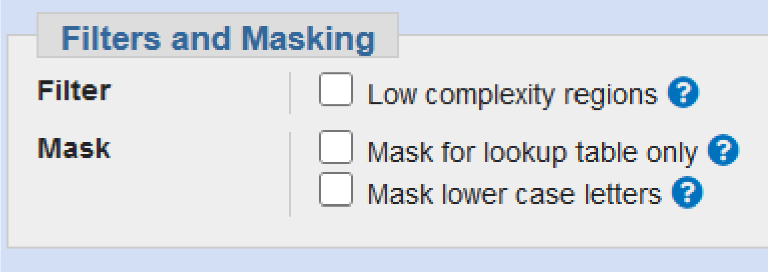
Low-Complexity Regions

Low-Complexity Regions I

- When working with Dotplots you could produce plots with large dark squares indicating so-called low-complexity regions
- These regions can consist of:
  - the same Amino acid, e.g. AAAAAAAAAAAAAAA
  - a repeating pattern, e.g. CGCGCGCGCGCG
  - an overabundance of a certain amino acid, e.g. SSSSCGSSRRSSDSSS

Low-Complexity Regions II

- These Low-complexity regions can be problematic when using BLAST or other database search Tools
- Therefore BLAST allows you to mask these regions (image)
- If you do so, parts of your query sequence aren‘t taken into consideration during the search
- Which regions were masked can be looked up in the optimal Alignments, since the query sequence is written in lower case at these parts



Better Protein BLASTs

Better Protein BLASTs I – Limits of BLAST

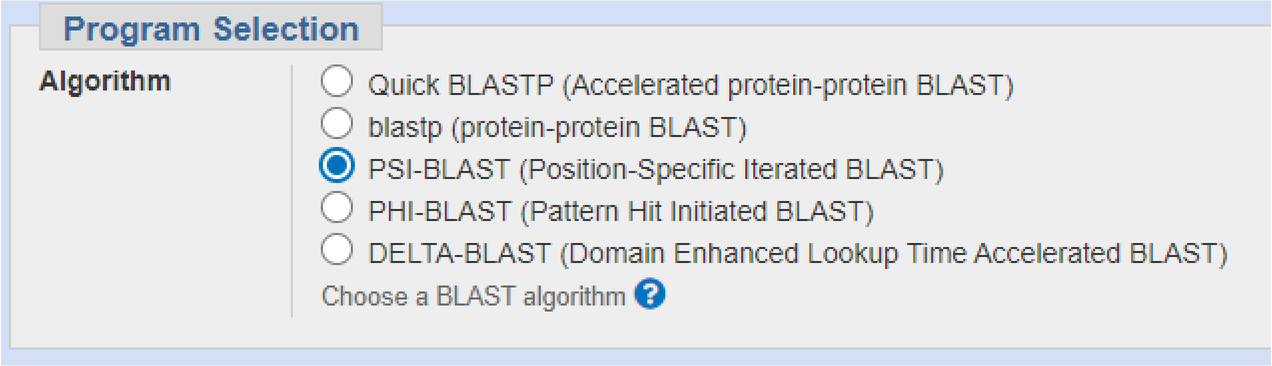
- The normal Protein BLAST reaches its limits when we are trying to detect distant relationships in the realm of identities around 30%
- To be able to find even such relationships some improved Protein BLASTs were developed
- These work with the general BLAST algorithm but have some additional features to improve the quality of the results

Better Protein BLASTs II – PSI-BLAST I [13]

- **P**osition-**S**pecific **I**terative BLAST
- Derives a position-specific scoring matrix (PSSM) from the multiple sequence alignment of sequences detected above a given score threshold using protein–protein BLAST
- This PSSM is used to further search the database for new matches, and is updated for subsequent iterations with these newly detected sequences
- Provides a means of detecting distant relationships between proteins

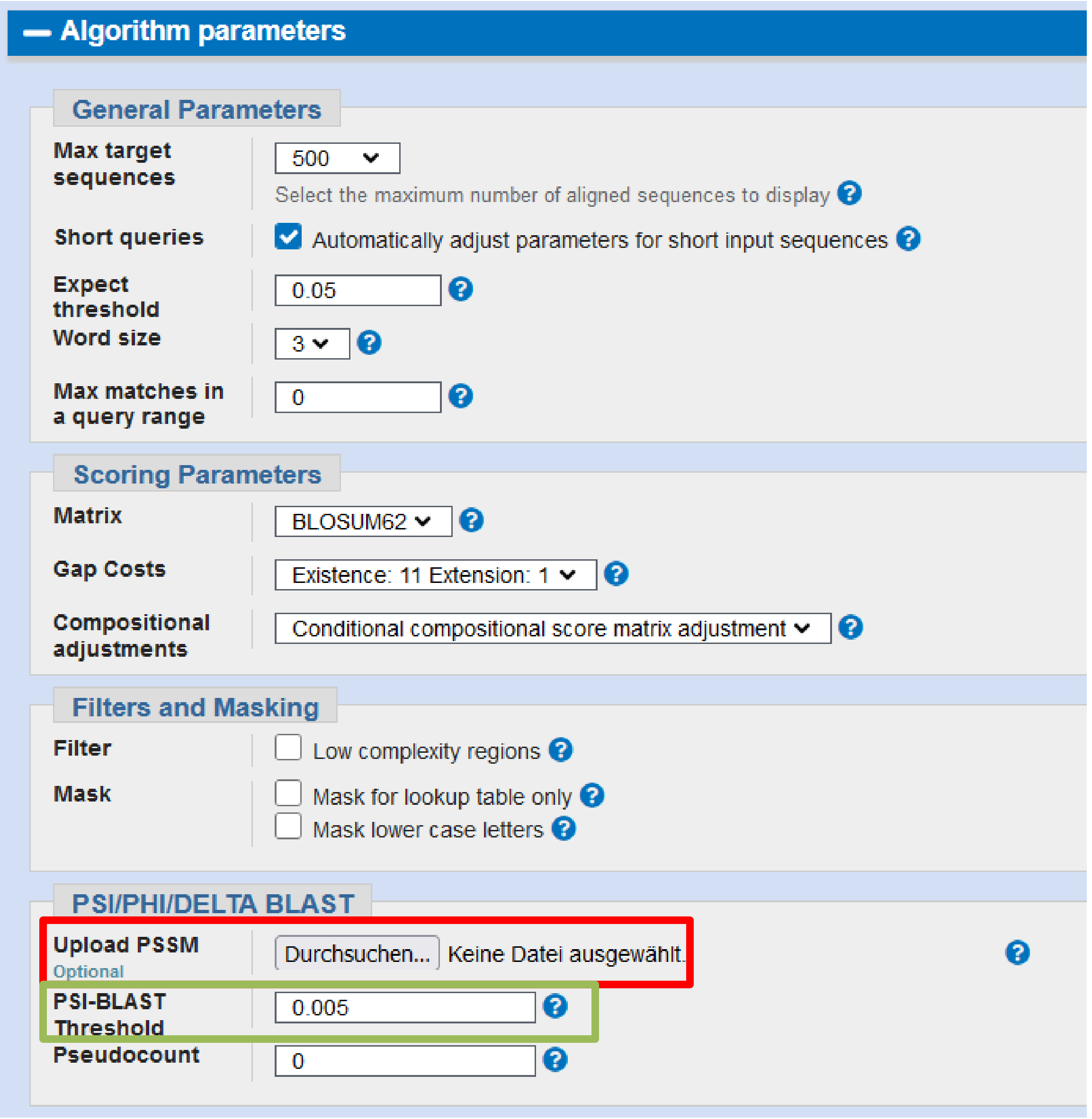
Better Protein BLASTs III – PSI-BLAST II [13]

- The general approach when using a PSI-BLAST is to run a Protein BLAST with the activated option of PSI-BLAST first and use the results of this run to calculate the PSSM (see next slides for more information)
- Afterwards you restart the BLAST search to look for additional results
- In most cases 2 to 5 iterations using a PSSM should be enough to find a good amount of potential results



Better Protein BLASTs IV – PSI-BLAST III [13]

- When working with PSI-BLAST we have some more parameters available to us before we start the first search
- Especially the upload of an already existing PSSM (red) as well as the adjustment of the threshold to determine whether a newly found sequence is relevant or not (green) are useful



Better Protein BLASTs V – PSI-BLAST IV [13]

- After the first search you can start the second iteration on the result page (left)
- After the second search new results are marked in the table (bottom)

hypothetical protein [Bacteroides sp. AF35-22]	Bacteroides sp. A...	474	474	97%	3e-156	26.22%	656	WP_120046740.1
conserved hypothetical protein [uncultured Dysgonomonas sp.]	uncultured Dysgo...	472	543	93%	3e-156	34.54%	603	SBW06303.1
hypothetical protein [Bacteroides sp. AM32-11AC]	Bacteroides sp. A...	474	474	97%	4e-156	26.22%	656	WP_119957683.1
hypothetical protein [Bacteroides]	Bacteroides	473	473	97%	4e-156	26.22%	656	WP_034529142.1
hypothetical protein [Phocaeicola sartorii]	Phocaeicola sartorii	467	467	72%	5e-156	38.21%	477	NBH64855.1
hypothetical protein [Bacteroides]	Bacteroides	473	473	97%	5e-156	26.22%	656	WP_005824411.1
hypothetical protein [Bacteroides uniformis]	Bacteroides unifor...	473	473	97%	8e-156	26.22%	656	WP_117866462.1
hypothetical protein [Bacteroides uniformis]	Bacteroides unifor...	473	473	97%	8e-156	26.29%	656	WP_151858663.1
hypothetical protein [Bacteroides uniformis]	Bacteroides unifor...	473	473	97%	9e-156	26.22%	656	WP_057087462.1
hypothetical protein [Bacteroides]	Bacteroides	473	473	97%	9e-156	26.22%	656	WP_087332889.1
hypothetical protein [Bacteroides uniformis]	Bacteroides unifor...	472	472	97%	1e-155	26.22%	656	WP_151875823.1
hypothetical protein [Bacteroides uniformis]	Bacteroides unifor...	472	472	97%	1e-155	26.44%	656	WP_151854945.1

Better Protein BLASTs VI – PSI-BLAST V – PSSM I [14]

- Are often derived from a set of aligned sequences that are thought to be functionally related
- Has one row for each symbol of the alphabet (4 rows for nucleotides or 20 rows for amino acids) and one column for each position in the pattern / Alignment
- First a basic position frequency matrix (PFM) is created by counting the occurrences of each letter at each position
- From the PFM, a position probability matrix (PPM) can now be created by dividing that former letter count at each position by the number of sequences, thereby normalizing the values

Better Protein BLASTs VII – PSI-BLAST VI – PSSM II [14]

- Most often the elements in PSSMs are calculated as log likelihoods
- That means the elements of a PPM (M(k,j)) are transformed using a background model b:
  - $M(k, j) = \log_2(M(k, j) / b_k)$
  - Describes how an element in the PSSM (left) can be calculated by using the values of the PPM (right)
- The simplest background model assumes that each letter appears equally frequently in the dataset:
  - $b_k = 1/|k|$ with k=number_of_letters_in_alphabet
  - 0.25 for nucleotides and 0.05 for amino acids

Better Protein BLASTs VIII – PSI-BLAST VII – PSSM III [14]

GAGGTAAAC

TCCGTAAGT

CAGGTTGGA

ACAGTCAGT

TAGGTCATT

TAGGTACTG

ATGGTAACT

CAGGTATAC

TGTGTGAGT

AAGGTAAGT

The corresponding PFM is:

```
A [ 3 6 1 0  0  6 7 2 1 ]
C [ 2 2 1 0  0  2 1 1 2 ]
G [ 1 1 7 10 0  1 1 5 1 ]
T [ 4 1 1 0  10 1 1 2 6 ]
```

Therefore, the resulting PPM is:

```
A [ 0.3 0.6 0.1 0.0 0.0 0.6 0.7 0.2 0.1 ]
C [ 0.2 0.2 0.1 0.0 0.0 0.2 0.1 0.1 0.2 ]
G [ 0.1 0.1 0.7 1.0 0.0 0.1 0.1 0.5 0.1 ]
T [ 0.4 0.1 0.1 0.0 1.0 0.1 0.1 0.2 0.6 ]
```

Applying the background model b to the PPM gives:

```
A [  0.26  1.26 -1.32 -∞   -∞    1.26  1.49 -0.32 -1.32 ]
C [ -0.32 -0.32 -1.32 -∞   -∞   -0.32 -1.32 -1.32 -0.32 ]
G [ -1.32 -1.32  1.49 2.0 -∞   -1.32 -1.32  1.0  -1.32 ]
T [  0.68 -1.32 -1.32 -∞   2.0 -1.32 -1.32 -0.32  1.26 ]
```

Better Protein BLASTs X – PHI-BLAST I [15]

- **P**attern-**H**it **I**nitiated BLAST
- Takes as input both a protein sequence and a pattern of interest that is part of your query sequence
- Searches a protein database for other instances of the input pattern, and uses those found as seeds for the construction of local alignments to the query sequence
- In many instances, the program is able to detect statistically significant similarity between homologous proteins that are not recognizably related using traditional single-pass database search methods.

Better Protein BLASTs XI – PHI-BLAST II – Patterns I [16]

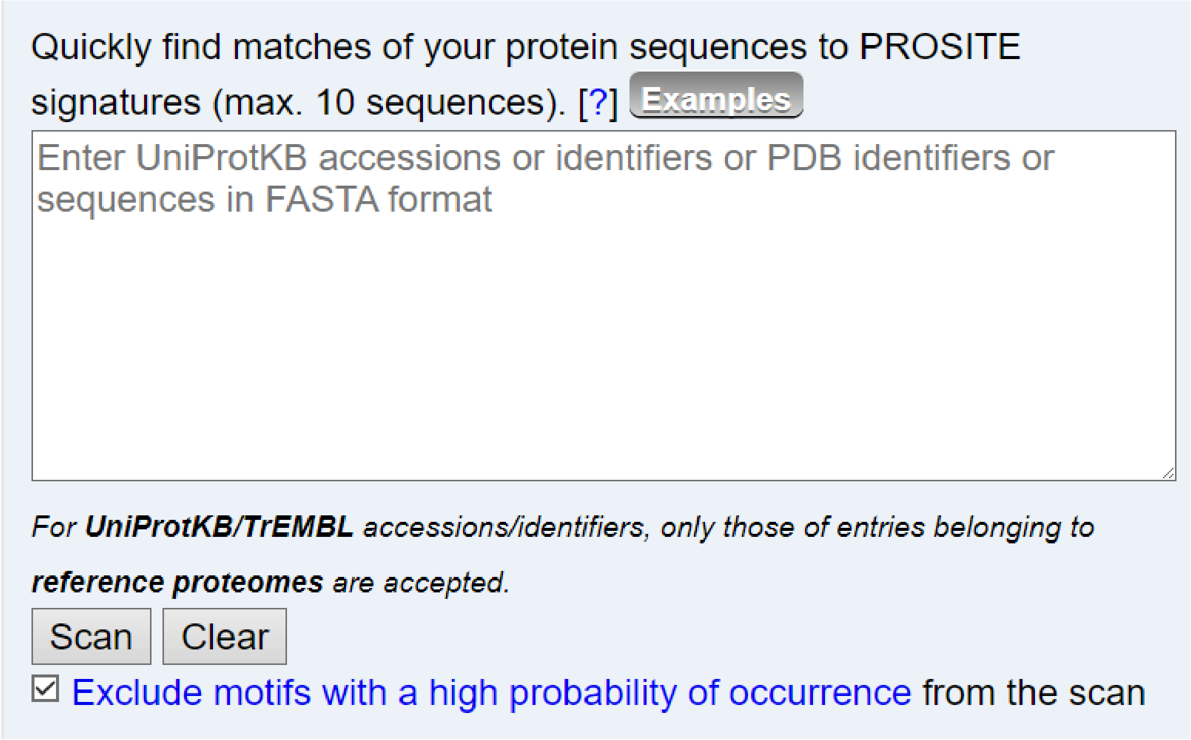
- PHI – Patterns allow us to search for regions that follow specific rules
- Example:

AA [DEQN] x(2,5) {YFWH}

  - First we have two Alanines
  - Than we have either Asparagine, Aspartic Acid, Glutamine or Glutamic Acid
  - Than we have 2 up to 5 „random“ amino acids
  - Than we have everything but Tyrosine, Phenylalanine, Tryptophane or Histidine

Better Protein BLASTs XII – PHI-BLAST III – Patterns II [16]

- The patterns you just saw stem from the Website Prosite: https://prosite.expasy.org
- On the start page of this database you can find a scanning tool that allows you to search for specific motifs / domains in your sequence



Better Protein BLASTs XIII – PHI-BLAST IV – Patterns III [16]

- Once your Scan is finished you get a tabular overview of all found domains / motifs etc.
- In the most right column you can see the different patterns found for your sequence (red rectangle)

8 - 43: **score =** 17.303

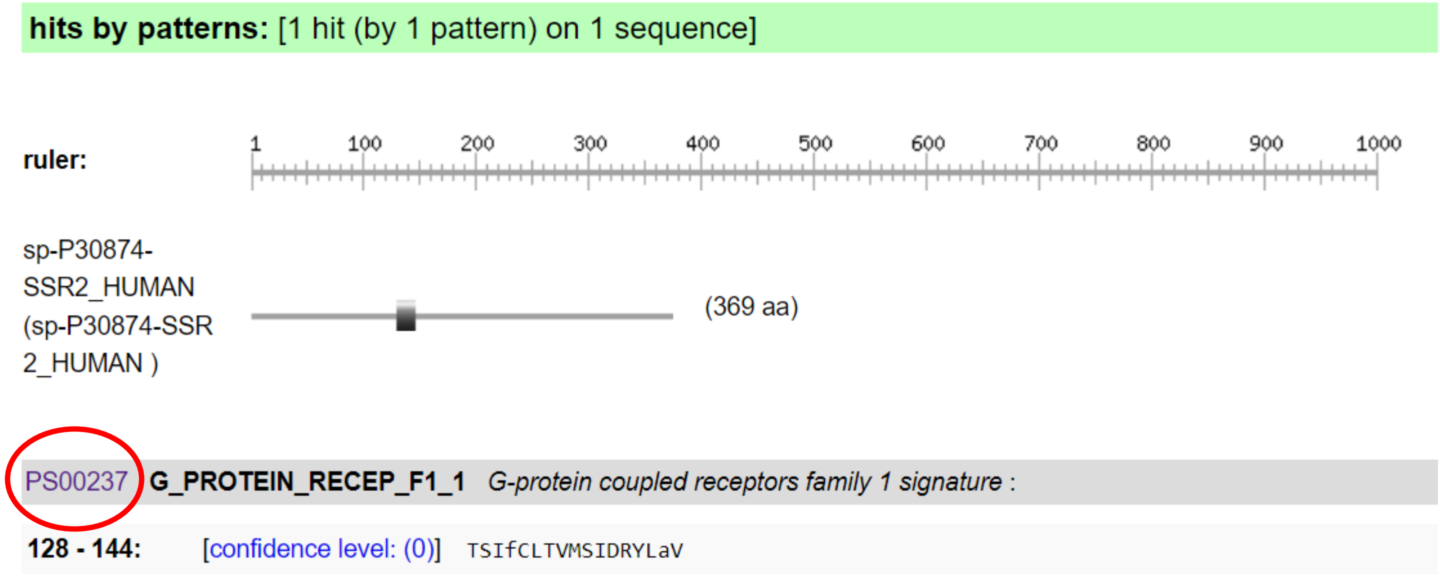
EQIAEFKEAFSLFDKDGDGTITTKELGTVMRsLGqN

Predicted features:

DOMAIN	8	43	EF-hand	[condition: none]
BINDING	21		/ligand="Ca(2+)" /ligand_id="ChEBI:CHEBI:29108" /ligand_label="1	[condition: [HCUEDKQNMYWSTRX] and <14-26=D-{W}-[DNS]-{ILVFYW}-[DENSTG]-[DNQGHRK]-{GP}-[LIVMC]-[DENQSTAGC]-x(2)-[DE]-[LIVMFYW]>]
BINDING	23		/ligand="Ca(2+)" /ligand_id="ChEBI:CHEBI:29108" /ligand_label="1	[condition: [HCUEDKQNMYWSTRX] and <14-26=D-{W}-[DNS]-{ILVFYW}-[DENSTG]-[DNQGHRK]-{GP}-[LIVMC]-[DENQSTAGC]-x(2)-[DE]-[LIVMFYW]>]

Better Protein BLASTs XIV – PHI-BLAST V – Patterns IV [16]

- Sometimes the ProSite Scan returns no patterns instead you will get results under “hits by patterns”
- In these cases you have to follow the link of the found pattern (red circle) to look up the pattern



Better Protein BLASTs XV – PHI-BLAST VI – Patterns V [16]

- On the site of the corresponding pattern you found, there can be a so called Technical section
- This section can contain a pattern for the corresponding entry in ProSite under Consensus pattern (red rectangle)

Technical section

PROSITE methods (with tools and information) covered by this documentation:

G_PROTEIN_RECEP_F1_1, PS00237; G-protein coupled receptors family 1 signature (PATTERN)

- Consensus pattern:
  [GSTALIVMFYWC]-[GSTANCPDE]-{EDPKRH}-x-{PQ}-[LIVMNQGA]-{RK}-{RK}-[LIVMFT]-[GSTANC]-[LIVMFYWSTAC]-[DENH]-R-[FYWCSH]-{PE}-x-[LIVM]

Better Protein BLASTs XVI – PHI-BLAST VII – Input

- When choosing PHI-BLAST under Algorithms an entry field appears on the web page that asks you to enter a PHI pattern (red rectangle)
- Be careful that the pattern you enter here can be found in your sequence

Better Protein BLASTs XVII – PHI-BLAST VIII – Error

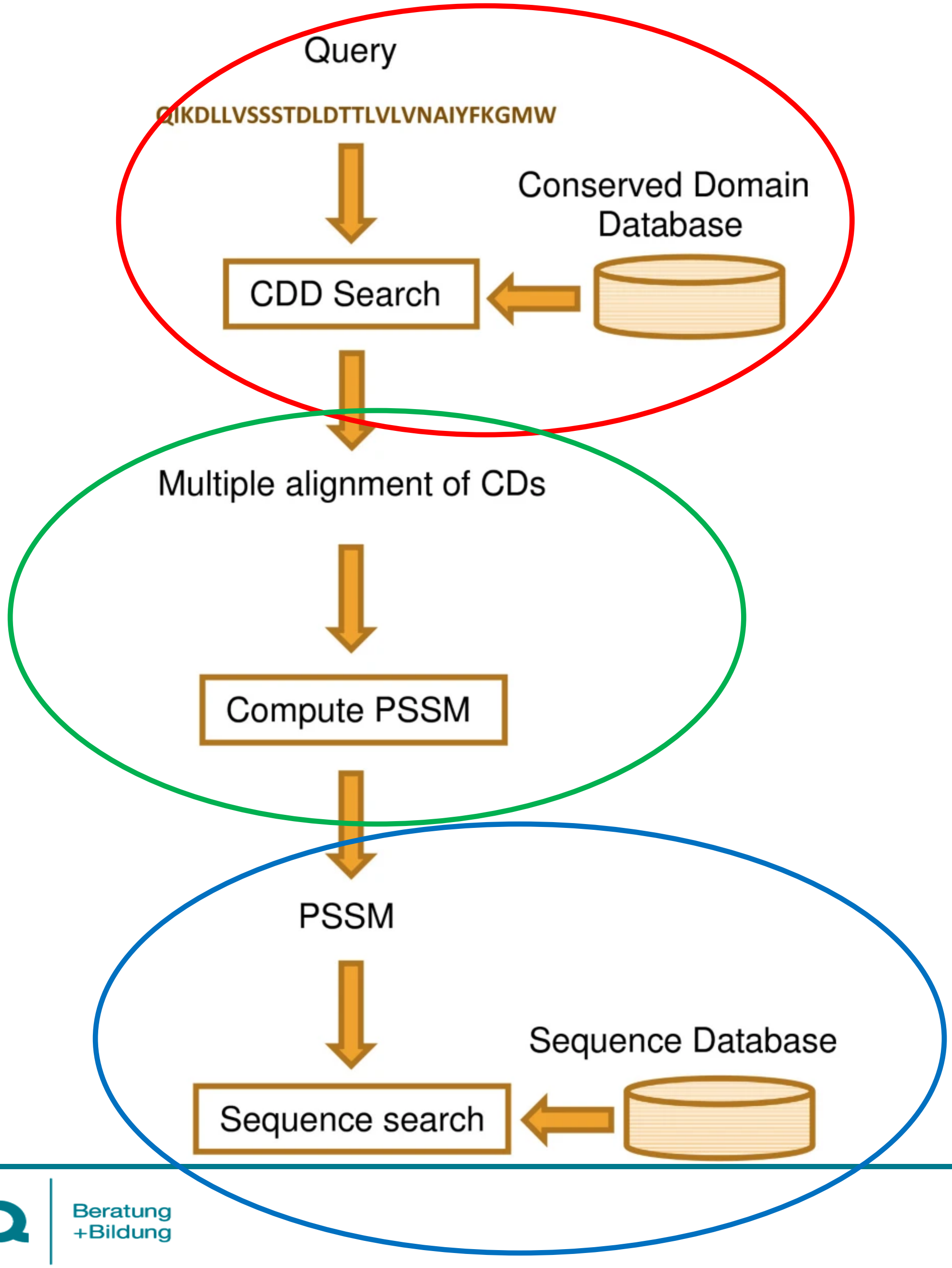
- If you try to run a PHI-BLAST with a pattern that is not present in your query sequence it won‘t work
- That means as soon as you get back an error message as shown below you tried to use PHI-BLAST with the wrong pattern

There was a problem with the search. Please, contact Help Desk and include RID 6Y421ERZ013.
Error: Warning: The pattern D-{W}-[DNS]-{ILVFYW}-[DENSTG]-[DNQGHRK]-{GP}-[LIVMC]-[DENQSTAGC]-X(2)-[DE]-[LIVMFYW]>] was not found in the query.

Better Protein BLASTs XVIII – Delta BLAST I [17]

- **D**omain **E**nhanced **L**ookup **T**ime **A**ccelerated BLAST
- This version of the Protein BLAST executes a multi-step approach to get to better results
- In the first step it uses a RPS-BLAST [18] to search through the conserved domain database (CDD) to find domains that are similar to the search query
- Since the CDD contains MSAs of these domains we can not use a normal BLAST
- After DELTA BLAST identified similar domains it calculates a PSSM for our search query using these MSAs and than uses this PSSM in a BLAST search similar to PSI-BLAST

Better Protein BLASTs XIX – Delta BLAST II [17]



- On the left you can see the general approach of a DELTA BLAST
- First we run a RPS-BLAST in the CDD (red)
- Than we create a MSA and calculate a PSSM (green)
- Last but not least we use the PSSM for a PSI-BLAST (blue)

PSI-BLAST Example D2AGE2 (4 Iterations)

PHI-BLAST Example P98088 (negative)
PHI-BLAST Example A0A024V8G6 (positive? / ProSite Profiles)

DELTA-BLAST Example D2AGE2

Exercise 11

Exercise 12

Exercise 13

Exercise 14

Exercise 16

Exercise 15: Other Protein Blasts

- Run BLAST searches using the same sequences as in Exercise 1
- Start multiple Runs using the different search parameters shown in the table on the right
- Please change only the mentioned parameters

Run	Type	Parameters	Alignments
Run 1	PSI	Default	500
Run 2	PSI	BLOSUM45	500
Run 3	PSI	PAM250	500
Run 4	PSI	PAM30	500
Run 5	PSI	word size 2	500
Run 6	PHI	Default	500
Run 7	PHI	BLOSUM45	500
Run 8	PHI	PAM250	500
Run 9	PHI	PAM30	500
Run 10	PHI	word size 2	500
Run 11	DELTA	Default	500
Run 12	DELTA	BLOSUM45	500
Run 13	DELTA	PAM250	500
Run 14	DELTA	PAM30	500
Run 15	DELTA	word size 2	500

Literature I

1. https://web.expasy.org/docs/relnotes/relstat.html (Release: 05_2022, Last visited: 19.01.2023)
2. https://en.wikipedia.org/wiki/FASTA (Last edited: 20.11.2022, Last visited: 19.01.2023)
3. Lipman, D J, and W R Pearson. “Rapid and sensitive protein similarity searches.” Science (New York, N.Y.) vol. 227,4693 (1985): 1435-41. doi:10.1126/science.2983426
4. https://www.spektrum.de/lexikon/biologie/sequenz-alignment/61121 (Last visited: 19.01.2023)
5. https://thebiologynotes.com/fasta/ (Last edited: 10.03.2023, Last visited: 19.05.2023)
6. https://en.wikipedia.org/wiki/Standard_score (Last edited: 29.03.2023, Last visited: 22.05.2023)
7. Pearson, W R. “Comparison of methods for searching protein sequence databases.” Protein science : a publication of the Protein Society vol. 4,6 (1995): 1145-60. doi:10.1002/pro.5560040613
8. https://www.ncbi.nlm.nih.gov/BLAST/tutorial/Altschul-1.html (Last visited: 19.01.2023)
9. https://resources.qiagenbioinformatics.com/manuals/clcgenomicsworkbench/650/_E_value.html (Last visted: 19.05.2023)
10. https://en.wikipedia.org/wiki/BLAST_(biotechnology) (Last edited: 05.05.2023, Last visted: 22.05.2023)

Literature II

10. Altschul, S F et al. “Gapped BLAST and PSI-BLAST: a new generation of protein database search programs.” Nucleic acids research vol. 25,17 (1997): 3389-402. doi:10.1093/nar/25.17.3389
11. https://en.wikipedia.org/wiki/Reading_frame (Last edited: 17.03.2022, Last visited: 19.01.2023)
12. https://www.oreilly.com/library/view/basic-applied-bioinformatics/9781119244332/c16.xhtml (Last visited: 19.01.2023)
13. Comparative Genomics: Volumes 1 and 2., Bergman NH, editor. Totowa (NJ): Humana Press; 2007. Chapter 10
14. https://en.wikipedia.org/wiki/Position_weight_matrix (Last edited: 08.11.2022, Last visited: 19.01.2023)
15. Zheng Zhang, Webb Miller, Alejandro A. Schäffer, Thomas L. Madden, David J. Lipman, Eugene V. Koonin, Stephen F. Altschul, Protein sequence similarity searches using patterns as seeds, Nucleic Acids Research, Volume 26, Issue 17, 1 September 1998, Pages 3986–3990
16. https://www.ncbi.nlm.nih.gov/blast/html/PHIsyntax.html (Last edited: 10.10.2000, Last visited: 19.01.2023)

Literature III

17. Boratyn, G.M., Schäffer, A.A., Agarwala, R. et al. Domain enhanced lookup time accelerated BLAST. Biol Direct 7, 12 (2012)
18. https://gensoft.pasteur.fr/docs/blast/2.2.26/rpsblast.html (Last visited: 25.05.2023)